

Arash Ahmadinia

Design and Implementation of a Smart Insole for the Prevention of Diabetic Foot Ulcers

Metropolia University of Applied Sciences
Master of Engineering, Information Technology
Medical Technology
Master's Thesis
May 2026

Abstract

Author:	Arash Ahmadinia
Title:	Design and Implementation of a Smart Insole for the Prevention of Diabetic Foot Ulcers
Number of Pages:	52 pages + 5 appendices
Date:	May 2026
Degree:	Master of Engineering
Degree Programme:	Information Technology
Specialisation option:	Medical Technology
Instructor(s):	Sakari Lukkarinen, Senior Lecturer Mikael Soini, Principal Lecturer

---

Diabetic foot ulcers are still a major medical problem, and most current solutions cannot monitor foot pressure in real time and reduce pressure together in one device. This study aimed to solve this problem by developing a smart insole that continuously measures plantar pressure and adjusts pressure distribution while walking. Using design science and a mixed-methods approach, the researcher combined expert feedback with practical testing during the development process. The final prototype included built-in pressure sensors and a manually adaptive pressure offloading for real-time pressure relief. Expert evaluations showed that the device is practical, clinically useful, and has strong potential for early intervention and more personalized diabetic foot care. This study suggests that a user-centered and cost-effective smart insole can help prevent and manage diabetic foot complications in the long term.

Keywords: Smart Insoles, Diabetic Foot Ulcers, Load Offloading Mechanism, Configurable Modular Architecture, Wearable Medical Device

---

The originality of this thesis has been checked using Turnitin Originality Check service.

Use of AI Tools

In accordance with the guidelines of Metropolia University of Applied Sciences regarding academic integrity and the broader principles of scientific transparency, the author declares the following use of artificial intelligence (AI) tools during the preparation of this thesis.

I used version 4 of OpenAI's ChatGPT to support the planning and structuring of the thesis, as well as to create images based on text prompts. I also used the tool for language refinement and citation formatting. As the author of this thesis, I take full responsibility for all of its content.

AI was not used to generate research findings, formulate the research problem, conduct an independent literature review, analyse interviews, or generate academic ideas. All referenced sources were identified, read, and evaluated by the author. The research design, architectural decisions, prototype design, and academic arguments presented in this thesis are entirely the author's original intellectual contribution.

The author accepts full responsibility for the content, accuracy, and validity of this thesis, and confirms that the use of AI tools described above complies with the academic integrity policies of Metropolia University of Applied Sciences.

Contents

1 Introduction 1

2 Background literature 6

2.1 Stages of diabetic foot ulcers 6
2.2 Current Trends and Technological Integration 11
2.2.1 Pressure Offloading Insoles 11
2.2.2 Smart Monitoring Insoles 12
2.3 Clinical Need and Rationale for Smart Insoles 13
2.4 Clinical Evidence Supporting Smart Insoles 14
2.5 Alternative Treatments and Comparative Analysis 15
2.6 Market Overview 16
2.7 Market Size and Growth Potential 19

3 Patent Application Process 20

3.1 Patentability Criteria 22
3.2 Non-Patentable Subject Matter 22
3.3 Components of a Patent Application 23
3.4 Freedom to Operate (FTO) 24
3.5 Search and Documentation 24
3.6 Timeline of the Process 25
3.7 Summary of Patentability 26

4 Research Ethics and Research Permit Process 29

4.1 Research Permission Requirements 29
4.2 Preparation of Research Documentation 31
4.3 Research Permit Application Process atMetropolia 33
4.4 Research Permit Application Process at City of Helsinki 34
4.5 Conclusion 35

5 Methods and Materials 35

5.1 Participants and Sampling 37
5.2 Data Collection and Analysis 38

6 Artifact Description 38

6.1 CRM-Based Design Process and Concept Development 40

6.2 Design Principles 42
6.3 System Architecture 43
6.4 Technical Implementation 44
6.5 Prototype Design Description 45
6.6 Evaluation 46

7 Discussion and Conclusions 47

References 49

Appendices
Appendix 1 - Research Plan
Appendix 2 - Privacy Notice
Appendix 3 - Consent to Participate in Research
Appendix 4 - Information Sheet for Participants
Appendix 5 - Data Management Plan created using DMPTuuli

List of Abbreviations

ADA	American Diabetes Association
CAGR	Compound Annual Growth Rate
CRM	Conceptual Research Method
DFU	Diabetic Foot Ulcer
DFUs	Diabetic Foot Ulcers
DSR	Design Science Research
EPC	European Patent Convention
EPO	European Patent Office
EU	MDR European Union Medical Device Regulation
FDA	Food and Drug Administration
FTO	Freedom to Operate
GDPR	General Data Protection Regulation
HUS	Helsinki University Hospital
IDF	International Diabetes Federation
IMU	Inertial Measurement Unit
IWGDF	International Working Group on the Diabetic Foot
MPP	Maximum Plantar Pressure
MUAS	Metropolia University of Applied Sciences
MSOLE	Modular Smart Offloading Insole
NICE	National Institute for Health and Care Excellence
PAD	Peripheral Arterial Disease
PRH	Finnish Patent and Registration Office
PTI	Pressure Time Index
TCC	Total Contact Casting

1 Introduction

Diabetic Foot Ulcers (DFU) rank among the most severe chronic complications individuals living with diabetes face. They mainly happen due to nerve injury and inadequate circulation mess with feeling in the feet. If you lose that protective feeling, you simply don’t detect minor cuts, pressure points, or changes in your skin. Damage builds up quietly, so by the time doctors spot the ulcers, things usually aren’t good. At that point, the risk for infections, deeper tissue issues, and sometimes even amputation jumps. These sores significantly impact people's lives and put pressure on healthcare systems worldwide [1].

From a biomechanical perspective, abnormal plantar pressure distribution is a key contributing factor in ulcer formation. High-pressure regions, particularly under the heel and metatarsal heads, are exposed to repetitive mechanical stress during walking. In addition, structural deformities and altered gait patterns can further intensify localized pressure, accelerating tissue breakdown [2]. As a result, monitoring and managing plantar pressure has become a central strategy in the prevention and management of diabetic foot complications.

Various clinical interventions have been developed to reduce plantar pressure, including therapeutic footwear, orthotic devices, and specialized insoles. These approaches aim to redistribute loads and reduce peak pressures in high-risk areas [3]. In recent years, advances in sensing and wearable technologies have enabled the development of more sophisticated systems capable of providing improved pressure monitoring and personalized interventions [4]. Despite these advancements, several challenges remain, including limitations in pressure offloading efficiency, user comfort, and the time and cost associated with customization.

Most existing solutions rely on passive materials or require manual adjustments, which limits their ability to adapt to dynamic changes during gait and daily activities [5]. As highlighted in prior work, conventional orthotic systems often

provide static support and cannot effectively respond to variations in biomechanical conditions over time. This limitation underscores the need for more advanced and adaptive approaches that can continuously adjust to user-specific conditions in real time while remaining practical for everyday use.

Addressing these challenges is essential for reducing the incidence of severe complications, including amputation, and for improving long-term clinical outcomes and quality of life in individuals living with diabetes [6].

The aim of this study is to enhance the efficiency and improve the design of smart insoles for the prevention and/or treatment of diabetic foot ulcers. Significant progress has already been made in terms of clinical testing, usability, and overall insole performance. However, further research and design efforts are still required to specifically address the challenge of pressure offloading.

The main reason (DFUs) develop is repeated pressure or mechanical stress on areas of the foot that no longer feel pain due to nerve damage. Since people with sensory neuropathy cannot sense pain properly, these wounds often go unnoticed until they become serious. That’s why one of the most important steps in early DFU care is “pressure offloading,” which means using insoles or special devices to take pressure off the affected area [7].

Research shows that this really works. For instance, about 69% of studies report that insoles with an arch-shaped design increase the contact area of the foot and lower the pressure on vulnerable spots. In fact, arch supports alone reduced average plantar pressure by around 37.5 kPa (with 95% confidence, $p < 0.03$). By spreading the load more evenly, these designs reduce stress on high-risk areas, especially under the forefoot [8].

Other solutions like multi-density insoles, soft pads placed under the metatarsal region, and shoes with rocker soles have also been shown to cut down pressure and lower the chance of ulcers coming back [9]. Taken together, the evidence suggests that the best results come from a combination of methods that are

tailored to the patient. When biomechanics, personal needs, and new technology are considered together, DFU care becomes much more effective.

Personalized offloading is becoming especially important. Instead of depending on one standard device, the focus is shifting toward finding the right balance between reducing pressure, lowering shear forces, managing walking activity, and making sure patients actually use and accept the devices. Looking ahead, developments such as modular designs, advanced manufacturing techniques, and smart sensors will allow clinicians to adapt insoles and footwear to each patient's healing process [10].

Another review study mentioned that besides looking at the wider community, it is also important to focus on the personal needs of patients with diabetic foot ulcers. These needs can be met by using devices that are better in terms of cost, availability, and technical features [11].

However, there are many factors that require careful consideration. For instance, the durability of pressure sensors in smart insoles is important, because the bending of the shoe while walking and friction from the foot exposes them to significant wear and tear. Replacing the insole entirely would increase treatment costs. Another factor to consider is integrating any in-shoe plantar pressure measurement system, which requires balancing the available space inside the shoe with the space occupied by the device [12].

The primary objective of this study is to design and develop a user-centered smart insole capable of supporting the prevention and management of diabetic foot ulcers through continuous pressure monitoring and adaptive offloading. More specifically, the research aims to investigate how modularity, sensing technologies, and personalized pressure redistribution strategies can be integrated into a wearable medical device that is both clinically meaningful and practically usable in daily life. The study also seeks to contribute to the growing field of preventive healthcare technologies by proposing a cost-effective and

scalable solution that bridges the gap between traditional passive orthotics and modern smart monitoring systems.

Accordingly, the following research questions guide this study:

• How can a smart insole with pressure offloading capabilities be designed to support the prevention of diabetic foot ulcers?
This question focuses on the overall system architecture, integration of sensing technologies, and the practical feasibility of combining monitoring and offloading within a single device.

• How can modularity improve personalization and adaptability in smart insole systems?
This question investigates how interchangeable and configurable components can support individualized treatment and respond to different patient needs and biomechanical conditions.

• What technical and clinical factors influence the usability and effectiveness of the MSOLE system?
This question examines issues such as sensor placement, comfort, durability, usability, pressure redistribution performance, and clinician interaction with the system.

• What limitations and challenges may arise during the development and implementation of a modular smart offloading insole?
This question addresses practical constraints related to manufacturing, sensor reliability, maintenance, energy efficiency, and long-term clinical application.

This study follows a Design Science Research (DSR) framework combined with the Conceptual Research Method (CRM) to develop and evaluate a novel smart insole solution for diabetic foot ulcer prevention. The methodological approach was selected because the research focuses not only on understanding a clinical problem, but also on designing a practical and innovative healthcare artifact.

Unlike traditional experimental studies that primarily test predefined hypotheses, DSR supports iterative development, evaluation, and refinement of technological solutions in real-world healthcare contexts. At the same time, CRM enables the transformation of qualitative expert knowledge into structured design concepts and functional system requirements. Together, these methods provide a systematic pathway from problem identification to prototype development and evaluation.

The research process was conducted in several stages. First, a comprehensive literature review was performed to identify the current limitations of smart insoles, pressure offloading systems, and wearable medical technologies related to diabetic foot care. Following this phase, semi-structured interviews were conducted with experts from Metropolia University of Applied Sciences and VTT Technical Research Centre of Finland. The participants included specialists in orthotics, wearable sensors, rehabilitation technologies, and medical device development. Their insights were analyzed through coding and conceptual mapping techniques in order to identify recurring themes, technical requirements, usability concerns, and clinical expectations. These findings directly guided the development of the Modular Smart Offloading Insole (MSOLE).

The technical implementation phase focused on integrating plantar pressure sensing, modular architecture, and adaptive pressure offloading into a single wearable system. The prototype was designed with interchangeable sections corresponding to different anatomical regions of the foot, including the forefoot, midfoot, and rearfoot. Embedded pressure sensors continuously monitored plantar loading conditions, while the modular structure enabled targeted pressure redistribution based on identified high-risk areas. The system architecture was developed with a strong emphasis on usability, flexibility, low-cost implementation, and compatibility with daily footwear. In addition, preliminary functional testing and expert-based usability evaluations were performed to assess the feasibility, practicality, and clinical relevance of the proposed design.

2 Background literature

The management of (DFUs) has long prioritized pressure offloading as a critical strategy for both prevention and treatment. Initial interventions during the early stages of clinical management involved relatively simple techniques, including specialized padding and shoe modifications aimed at redistributing plantar pressure and minimizing mechanical stress on vulnerable foot regions [13].

2.1 Stages of diabetic foot ulcers

DFUs develop gradually through a series of progressive stages, beginning with minor skin abnormalities and potentially advancing to severe tissue destruction and gangrene if left untreated. These complications are mainly caused by peripheral neuropathy, poor blood circulation, repeated mechanical pressure, and delayed wound healing associated with diabetes. Early identification of ulcer progression is extremely important because timely intervention can significantly reduce the risk of infection, hospitalization, and lower-limb amputation. The figure below illustrates the commonly recognized stages of diabetic foot ulcer development, ranging from a high-risk foot without open wounds to severe gangrene involving the entire foot (Figure 1).

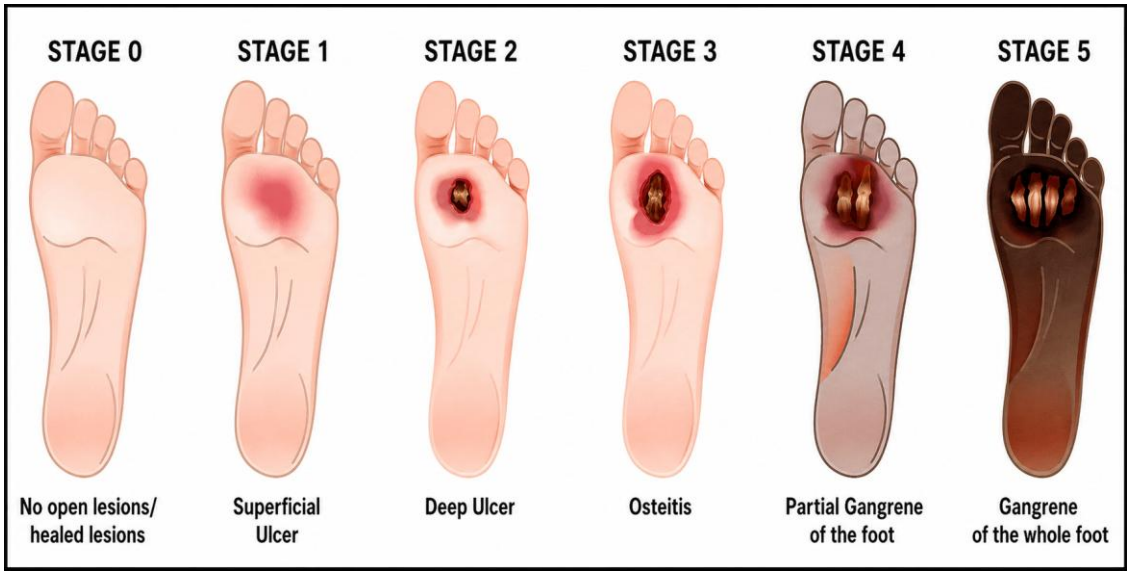


Figure 1. Stages of Diabetic foot ulcers [13]

Stage 0 – High-Risk Foot / Healed Lesions

At this stage, no open wound is present, but the foot is considered highly vulnerable to ulcer development. Patients may have calluses, dry skin, foot deformities, or previously healed ulcers. Peripheral neuropathy often reduces the ability to feel pain or pressure, increasing the risk of unnoticed tissue damage. Preventive care, regular foot inspection, pressure offloading, and proper footwear are essential at this stage to avoid ulcer formation [13].

Stage 1 - Superficial Ulcer

Stage 1 is characterized by a superficial ulcer that affects only the outer layer of the skin. The wound may appear as a small open sore surrounded by redness or inflammation. Although the damage is still limited, continuous pressure and poor blood circulation can quickly worsen the condition if treatment is delayed. Early clinical management at this stage usually includes wound cleaning, infection prevention, and pressure redistribution [13].

Stage 2 - Deep Ulcer

In Stage 2, the ulcer extends deeper into the soft tissues beneath the skin, potentially involving tendons, ligaments, or muscle tissue. The risk of infection becomes significantly higher because bacteria can penetrate deeper structures of the foot. Patients may experience swelling, discharge, and increased tissue damage around the wound area. More advanced wound care and specialized offloading interventions are often required to prevent further progression [13].

Stage 3 - Osteitis / Bone Involvement

Stage 3 indicates severe ulcer progression in which the infection reaches the bone, often resulting in osteitis or osteomyelitis. At this stage, the wound becomes deeper, more painful, and highly susceptible to serious infection. Pus formation, tissue destruction, and systemic infection may occur. Medical treatment becomes more aggressive and may include antibiotics, surgical debridement, and intensive pressure offloading strategies [13].

Stage 4 - Partial Gangrene

Stage 4 is characterized by localized gangrene affecting part of the foot, such as the toes or forefoot. Tissue death occurs due to severe infection and critically reduced blood supply. The affected area may appear dark, necrotic, and non-viable. At this stage, urgent medical intervention is necessary to stop the spread of tissue death and preserve as much healthy tissue as possible [13].

Stage 5 - Extensive Gangrene

Stage 5 represents the most severe stage of diabetic foot complications, where gangrene affects the entire foot. Extensive tissue necrosis and widespread infection create a life-threatening condition. In many cases, major surgical intervention or lower-limb amputation becomes unavoidable to prevent systemic infection and save the patient's life [13].

The literature search was carried out in a systematic way using several well-established scientific databases, including PubMed, Scopus, and Google Scholar. The search primarily focused on keywords: *Smart Insoles*, *Diabetic*, *Diabetic Foot Ulcers*, *Customized*, *Pressure Offloading*, and related technological innovations. In addition, studies were included if they explored complications associated with diabetes, the clinical outcomes of DFUs, manufacturing approaches for insoles, or commercially available products marketed as smart insoles for diabetic patients.

In total, the initial search yielded 5,449 peer-reviewed journal articles and conference papers with some level of relevance to smart insoles. This substantial number reflects the growing academic and clinical interest in the field, as well as its potential significance for both biomedical engineering and patient care.

Inclusion Criteria

To ensure the relevance and quality of the review, studies were included if they met one or more of the following conditions:

- Explicitly addressed diabetic foot ulcers as part of their scope.

- Focused on clinical trials, meta-analyses, or systematic reviews, as these provide stronger levels of evidence.
- Investigated personalized insoles or the customization of insole design for individual patients.
- Specialized in pressure offloading technologies designed for DFU prevention and management.

Applying these criteria narrowed the pool to 47 key studies selected for detailed analysis.

Exclusion Criteria

Studies not published in English were excluded to maintain consistency and accuracy in interpretation.

Search Results

The refined search revealed the following distribution across research topics (Figure 2):

- About 5,449 studies explored the general concept of smart insoles.
- Around 874 studies specifically examined diabetic.
- Roughly 495 studies focused on diabetic foot ulcers.
- About 81 studies investigated customized of insoles.
- Finally, 47 studies dealt directly with pressure offloading.

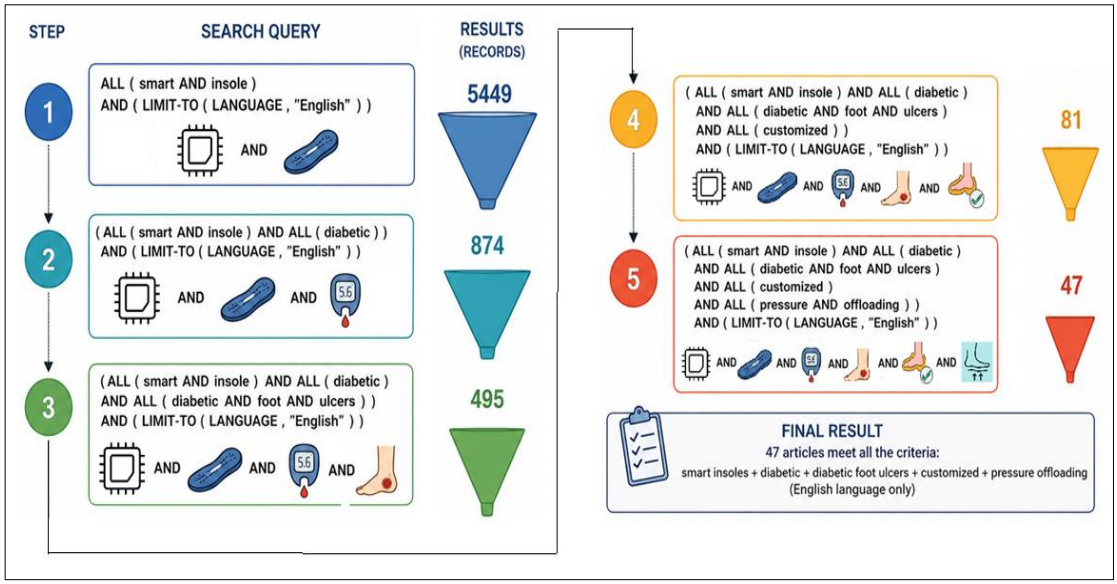


Figure 2. Flow of Literature Search and Selection Process

The development of diabetic foot complications (TCCs) usually follows a series of stages. It often starts with either healthy skin or completely healed wounds from previous ulcers. From there, a superficial ulcer may appear, affecting only the outer layers of the skin without reaching deeper tissues. As the condition worsens, the ulcer can extend further into the tissue, involving structures such as tendons, bones, or joints. When these deep ulcers become infected, the situation becomes more serious, leading to infected deep tissue wounds. If treatment is delayed, tissue death (necrosis) may occur, beginning with partial gangrene and, in the most severe cases, progressing to full gangrene.

A significant advancement came in the 20th century with the introduction of Total Contact Casting (TCC), which provided a standardized and highly effective means of pressure relief. This approach not only accelerated healing but also significantly reduced ulcer recurrence rates, establishing TCC as the gold standard for offloading in DFU treatment (Figure 3) [14].



Figure 3. Total Contact Casting (TCC) [14]

As research progressed into the late 20th and early 21st centuries, attention shifted toward custom orthotic solutions designed to accommodate individual patient anatomy. These personalized devices improved both comfort and pressure distribution while promoting greater patient compliance [5]. However, traditional offloading strategies despite their effectiveness faced persistent challenges, particularly in ensuring continuous patient adherence and providing real-time insight into plantar pressure dynamics.

2.2 Current Trends and Technological Integration

Diabetic insoles are generally classified into two main categories based on their primary function: pressure offloading insoles and smart monitoring insoles [15].

Pressure offloading insoles mainly focus on reducing excessive plantar pressure by redistributing body weight across the foot surface, thereby helping to prevent tissue breakdown and diabetic foot ulcer formation. These insoles are typically manufactured using soft cushioning materials and are often customized to match the patient's foot anatomy and gait characteristics. In contrast, smart monitoring insoles integrate wearable sensor technologies capable of continuously measuring plantar pressure, temperature, and gait-related parameters in real time. By collecting and transmitting data through wireless systems, these intelligent insoles provide early warnings of abnormal pressure conditions and enable timely clinical intervention. Together, both categories play an important role in diabetic foot care, combining mechanical pressure relief with advanced monitoring technologies to improve ulcer prevention, patient safety, and long-term foot health.

2.2.1 Pressure Offloading Insoles

Pressure offloading insoles are designed to relieve high-pressure areas on the foot by redistributing weight more evenly. They are specifically used to prevent and manage diabetic foot ulcers in people with diabetes. Research has shown that incorporating soft materials, such as PORON Medical 4708, into the insoles can significantly lower peak pressure (MPP) and the pressure-time index (PTI) on the forefoot and the dorsum of the foot. This reduction in pressure helps improve overall weight distribution and lowers the risk of developing diabetic ulcers. Furthermore, customized insoles have been found to be more effective in preventing diabetic foot ulcers than standard, off-the-shelf insoles [15].

Figure 4 shows three case studies of diabetic patients, each of whom visited the wound clinic with a form of diabetic foot ulcer. In each case, the initial condition

of the wound including its location and severity is documented visually, and the subsequent treatment process is shown using pressure offloading and the application of custom or amniotic membrane.

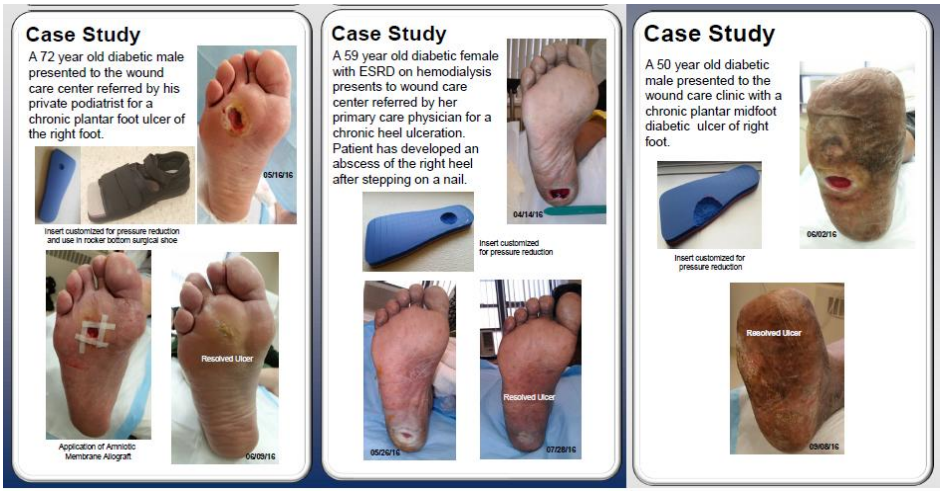


Figure 4. Clinical performance of pressure offloading insoles [16].

2.2.2 Smart Monitoring Insoles

In recent years, innovations in wearable technology have paved the way for real-time pressure monitoring solutions, notably through smart insoles equipped with pressure sensors and wireless connectivity. These systems allow for continuous data collection, enabling both patients and clinicians to identify and address high-risk pressure points before ulceration begins [17,18] and by offering immediate, personalized feedback, smart insoles enhance user engagement and support more precise offloading strategies tailored to daily activity and foot conditions.

Parallel advances in material science and 3D printing have further revolutionized offloading device design. New generation footwear and orthotic devices offer superior durability, biomechanical precision, and fit customization, contributing to more effective pressure redistribution and increased user comfort (Figure 5) [19].



Figure 5. Pedar-X plantar pressure measurement system [19]

Moreover, the integration of digital health interventions such as mobile apps that support patient self-care, footwear selection, and monitoring routines has shown promise in promoting education and behavioral change. These tools aim to reduce DFU risk by empowering patients to manage their condition proactively [20].

While Total Contact Casting (TCC) remains the benchmark for treating active ulcers, the integration of sensor-driven monitoring technologies, personalized orthotic solutions, and digital health tools is shaping a more preventive, patient-centered future in diabetic foot care.

2.3 Clinical Need and Rationale for Smart Insoles

Diabetic foot ulcers affect approximately 15–25% of diabetic patients during their lifetime and are a leading cause of non-traumatic lower-limb amputations globally. Peripheral neuropathy results in the loss of protective sensation, while abnormal plantar pressure increases the risk of tissue breakdown and ulcer formation. Impaired microcirculation further complicates healing and raises infection risk.

Intermittent clinical assessments (e.g., every 3-6 months) often miss the early signs of ulcer development, making continuous monitoring essential [21]. Traditional strategies such as total contact casting, custom orthotics, and foot

inspections are reactive rather than preventive, and often limited by patient compliance and episodic follow-ups.

Smart insoles are emerging as a solution to this unmet clinical need. By continuously monitoring plantar pressure, temperature, and even gait abnormalities, they deliver real-time alerts that prompt behavioral changes or clinical intervention. International guidelines from the International Diabetes Federation (IDF), American Diabetes Association (ADA), National Institute for Health and Care Excellence (NICE), and International Working Group on the Diabetic Foot (IWGDF) emphasize the importance of proactive strategies, including real-time pressure monitoring [22], [23], [24].

2.4 Clinical Evidence Supporting Smart Insoles

Clinical evidence shows that smart insoles can play an important role in caring for diabetic feet. They track foot pressure, temperature, and walking patterns, helping to spot problems before ulcers develop. Studies suggest that the real-time alerts and feedback these insoles provide can lower the risk of ulcers and help patients follow preventive measures. These results set the stage for clinical trials and reviews that further examine how well they work [25].

Clinical trials and systematic reviews provide strong evidence for smart insoles effectiveness, and 71% reduction in ulcer recurrence among smart insole users compared to controls [26]. The surrosense Rx® trial showed an 83-86% reduction in ulcer recurrence at high-risk plantar sites, with economic analysis indicating 70% reduction in prevention costs and long-term cost-effectiveness analysis indicated that using smart insoles as an adjunct to standard care reduced the cost per ulcer avoided to $6,700, compared to $53,100 with standard care alone [27].

2.5 Alternative Treatments and Comparative Analysis

Table 1 compares 3 strategies for stress remedy and foot care: TCC, custom offloading gadgets, and smart insoles with stress sensors. In phrases of cost, TCC and custom gadgets fall withinside the medium range, despite the fact that custom answers can grow to be extra steeply-priced because of individualized fabrication, even as smart insoles are the maximum luxurious due to embedded electronics and maintenance. When it involves accuracy, each TCC and smart insoles carry out pretty properly, TCC thru uniform redistribution of stress and smart insoles thru real-time measurement, even as custom gadgets also are powerful however rely closely on how properly they're designed and fitted. However, those advantages include trade-offs in usability: TCC is much less user-pleasant and restricts mobility significantly, while custom gadgets are simpler to apply and permit extra movement, and smart insoles take a seat down someplace in between.

Table 1. Comparative Table of Alternative Treatments [16],[17],[19].

Parameter	**TCC**	**Custom Offloading Devices**	**Smart Insoles with Pressure Sensors**
Cost	Medium	Medium to High (custom fabrication)	High (sensors, electronics, maintenance)
Accuracy in Pressure Relief	Very High (uniform redistribution)	High (depends on fit/design)	Very High (real-time direct measurement)
User-Friendliness	Low	High	Medium
Reliability	Very High (proven method)	High (fit-dependent durability)	Medium (sensor calibration/maintenance required)
Type of Pressure Measurement	Indirect (pressure offloading)	Indirect (custom molding)	Direct (real-time electronic pressure sensing)
Mobility Restriction	High	Low–Medium	Low
Need for Specialist	High	Medium	Low
Durability/Lifespan	Medium	High	Medium

In phrases of reliability and realistic use, TCC is taken into consideration very dependable because of its long-status scientific use, even as custom gadgets also are reliable however inspired through in shape and sturdiness over time. Smart insoles, despite the fact that technologically advanced, have mild reliability due to the fact they require calibration and ongoing maintenance. Another key distinction lies in how stress is measured: TCC and custom gadgets rely upon oblique methods (offloading and shaping), even as smart insoles offer direct, real-time data. Additionally, TCC calls for a consultant for correct application, custom gadgets want a few expert adjustment, and smart insoles are in general user-operated. Finally, sturdiness varies: custom gadgets remaining the longest with right care, TCC is temporary (normally used for weeks), and smart insoles have a mild lifespan because of sensor limitations.

2.6 Market Overview

The rapid advancement of wearable health technologies has significantly transformed the landscape of plantar pressure monitoring and foot care solutions. These improvements are especially critical in regions which includes diabetic foot care, rehabilitation, and gait analysis, wherein early detection of atypical stress styles can save from critical complications.

Orpyx® SI Sensory Insoles (Canada):
Provides real-time plantar pressure monitoring with alert-based feedback to help prevent excessive loading in high-risk areas. Primarily designed for diabetic foot management and early intervention [28].

Moticon® ReGo (Germany):
Offers fully integrated wireless insoles with embedded sensors for gait and pressure analysis. Suitable for both clinical use and everyday monitoring without external attachments [29].

FeetMe® Monitor (France):
Combines pressure sensing and motion tracking to analyse gait patterns and detect abnormalities. Widely used in rehabilitation and research environments [30].

Sensoria® Health (USA):
Develops smart textile-based insoles with real-time monitoring and mobile connectivity. Focuses on continuous tracking and user engagement in daily life [31].

PedarX Plantar System:
A high-precision plantar pressure measurement system commonly used in clinical and research settings. Provides detailed biomechanical data but is less suited for continuous everyday use [32].

Table 2 presents a comparative overview of commercially available smart insole and pressure offloading solutions related to diabetic foot care and gait monitoring. The comparison includes products such as si sensory insoles [28], moticon [29], feetme [30], sensoria [31], pedar-x [32], boogio [33], arion [34], salted [35], tg0 [36], nurvv [37], pedifix [38], and darco [39].

The analysis highlights a clear distinction between systems that primarily focus on smart monitoring capabilities and those designed mainly for passive pressure offloading. Most smart insole products provide features such as plantar pressure sensing, gait analysis, activity tracking, and real-time feedback, but they lack active pressure redistribution mechanisms. In contrast, traditional offloading devices mainly offer orthopedic support and pressure relief without embedded sensing or adaptive feedback technologies.

Table 2. Gap between DFU patient needs and market products [30].

Product Name and manufacturer	**Smart Monitoring Insoles**	**Pressure Offloading Insoles**	**Notes**
Orpyx® SI Sensory Insoles - Orpyx Medical Technologies [28].	✓	x	Have real-time pressure monitoring and alert feedback – no active pressure offloading
Moticon® - Moticon GmbH [29].	✓	x	Have wireless gait analysis and embedded sensors – no adaptive pressure control
FeetMe® - FeetMe SAS [30].	✓	x	Have motion tracking and clinical gait assessment – no pressure redistribution mechanism
Sensoria® - Sensoria Health Inc [31].	✓	x	Have smart textile sensing and mobile connectivity – no active biomechanical adjustment
Pedar-X - Novel GmbH [32].	✓	x	Have high-precision pressure measurement and research-grade analysis – no real-time wearable adaptation
Boogio - Boogio Inc [33].	✓	x	Have activity tracking and pressure sensing – no therapeutic offloading capability
ARION – ARION [34].	✓	x	Have running analytics and performance feedback – no medical pressure management
Salted - Salted Venture [35].	✓	x	Have AI-based motion analysis and balance tracking – no plantar pressure offloading
TG0 - TG0 Ltd [36].	✓	x	Have flexible sensing technology and integration capability – no standalone therapeutic function
Nurvv - Nurvv Ltd [37].	✓	x	Have detailed running metrics and real-time coaching – no clinical pressure redistribution
PediFix® - PediFix Inc [38].	x	✓	Have passive orthotic support and comfort solutions – no sensing or smart adaptation
DARCO® - DARCO International Inc [39].	x	✓	Have medical offloading devices and protective footwear – no real-time monitoring system
Current Research	✓	✓	Have integrated sensing and dynamic pressure offloading

This comparison demonstrates a significant gap in the current market, namely the absence of an integrated wearable system that combines both intelligent monitoring and dynamic pressure offloading. The “Current Research” row

addresses this limitation by proposing the MSOLE system, which integrates both functionalities into a single modular and personalized solution aimed at improving the prevention and treatment of diabetic foot ulcers through a more comprehensive, adaptive, and responsive design approach.

DFU prevention market especially smart insoles are undergoing rapid expansion, driven by global demographic trends, technological advancements, and healthcare policy shifts toward preventive care. Specialized companies with patented sensor technologies and clinical validation are leading the market [40].

Notable firms such as orpyx Medical Technologies and Sensoria Health have gained regulatory clearance and demonstrated clinical effectiveness, enabling broad commercialization. Market projections estimate a compound annual growth rate (CAGR) of 7–8% over the next five years, with smart insoles positioned as a central segment [28].

2.7 Market Size and Growth Potential

The global DFU prevention market, including smart insoles, offloading footwear, and advanced wound care, is projected to grow from USD 4.1 billion in 2023 to USD 5.8 billion by 2028, at a CAGR of ~7.5% [41]. Specifically, the smart insole segment is expected to grow from USD 200 million in 2024 to over USD 600 million by 2028, supported by:

- Increasing diabetes prevalence
- Public awareness of DFU risks
- Accelerated digital healthcare transformation
- Aging populations and investment in wearable medical devices [42].

Conventional (passive) insoles are generally classified as low-risk medical devices due to their purely mechanical function. In the European Union under the Medical Device Regulation (EU MDR), they are typically categorized as Class I, as they are non-invasive and do not include active components or measurement capabilities. Similarly, in the United States, such products are usually classified as Class I devices by the FDA and are often exempt from premarket notification

(510(k)), since they provide basic support and do not directly influence clinical decision-making or patient safety.

Initially, smart insoles designed for plantar pressure monitoring introduce sensing and data transmission capabilities, which increases their regulatory classification. In the EU, if the device is limited to measurement and does not perform diagnostic or therapeutic actions, it is generally considered a Class I active, non-invasive measuring device, as it collects data without directly affecting physiological conditions [43].

3 Patent Application Process

The patent application process in Finland follows a structured and well-defined pathway that guides applicants from the initial evaluation of their invention to the final granting and maintenance of patent rights [44]. It begins with assessing whether the invention meets key patentability criteria, followed by careful preparation of application documents and submission to the Finnish Patent and Registration Office (PRH) [45].

which evaluates applications through a series of legal and technical examinations. Initially, inventors must determine whether their invention satisfies the fundamental patentability requirements, including novelty, inventive step, and industrial applicability. After this assessment, a prior art search is typically conducted using international patent databases to identify existing technologies and avoid overlap with previously patented inventions. Once the uniqueness of the invention is confirmed, applicants prepare detailed technical documents, including descriptions, claims, figures, and abstracts that clearly define the scope and functionality of the invention. The application is then officially submitted to the PRH, which assigns a filing date and begins both formal and substantive examinations. During the examination stage, the authority may issue office actions requesting clarifications, modifications, or additional technical explanations, requiring the applicant to revise and resubmit parts of the application. After approximately 18 months, the application is published publicly,

allowing transparency and review within the patent system. If all legal and technical requirements are satisfied, the patent is granted, providing the inventor with exclusive commercial rights for up to 20 years, provided that annual maintenance fees are continuously paid to keep the patent active and enforceable.

Table 3. Patent Application Process in Finland: Step-by-Step Overview [45].

Step	Stage	Description
1	Patentability Assessment	Evaluate novelty, inventive step, and industrial applicability of the invention
2	Prior Art Search	Search existing patents using databases such as Google Patents and Espacenet
3	Technical Classification	Identify the appropriate technical field/class for the invention
4	Consultation	Seek advice from supervisors or experienced individuals in patent filing
5	Preparation of Documents	Prepare description, claims, drawings (if needed), and abstract
6	Filing the Application	Submit the application to the PRH and obtain a filing date
7	Examination	Formal and substantive review by the authority
8	Office Actions	Respond to objections or requests for clarification/amendments
9	Publication	Application is published after approximately 18 months
10	Granting of Patent	Patent is granted with exclusive rights (up to 20 years)
11	Maintenance Fees	Pay annual renewal fees to keep the patent valid
12	Post-Grant Management	Maintain or discontinue the patent based on its commercial value

Before submitting a patent application in Finland, it is essential to ensure that the invention meets the fundamental patentability criteria. These include novelty, inventive step, and industrial applicability. Novelty means that the invention must not have been disclosed to the public anywhere in the world before the filing date. The inventive step requires that the invention is not obvious to a person skilled in the relevant technical field.

3.1 Patentability Criteria

The first essential step for any researcher is ensuring the invention meets the legal criteria for a patent. In Finland and Europe, an invention must be novel, meaning it does not form part of the "state of the art". The state of the art includes everything made available to the public, whether through written documents, oral descriptions, or actual use, anywhere in the world before the filing date [45].

Beyond novelty, the invention must involve an inventive step, which means it must not be obvious to a person skilled in the specific technical field. Examiners often use a "problem-solution approach" to assess this, identifying the closest prior art and determining if the technical solution provided by the researcher would have been an obvious leap for an expert [45].

Finally, the invention must have industrial application, meaning it can be manufactured or used in any kind of industry. This ensures that patents are granted for practical, functional solutions rather than purely theoretical ideas. In the medical field, this is often demonstrated by showing the device can be used for diagnostic or therapeutic purposes within a clinical setting.

3.2 Non-Patentable Subject Matter

According to the European Patent Convention (EPC), certain categories of intellectual work are not considered inventions and are therefore excluded from patent protection and while a researcher may discover a new law of nature, that discovery itself cannot be patented; only a practical application of that discovery might qualify [45].

This means that schemes, rules, or methods for playing games, performing business activities, or purely mental exercises are not patentable and researchers should also note that computer programs as such are often excluded from patentability under these rules, though software that produces a ‘‘technical

effect ‘‘may sometimes be protected through other legal frameworks or specific patent types.

Understanding these exclusions early on helps a researcher avoid spending time and resources on filing applications for subject matter that the PRH or the EPO will automatically reject.

3.3 Components of a Patent Application

A formal patent application must begin with a clear Description of the invention. This written document must be so detailed and transparent that a person skilled in the same field can reproduce the invention based solely on the text. It typically starts with a factual title and must outline the technical problem the invention addresses, the proposed solution, and at least one detailed example of how the invention is applied in practice [46].

The Claims are the most critical part of the application because they define the exact scope of legal protection. Claims are generally divided into two parts: a preamble, which lists the features of the invention already known to the public, and a characterising part, which describes the new and inventive features. The phrase "characterised in that..." is standard terminology used to signal the start of the unique aspects being claimed for protection.

Lastly, the application must include an Abstract and Drawings where necessary. The abstract is a concise summary of no more than 150 words that explains the technical problem and the basic principle of the solution. Drawings are highly recommended because they help the examiner and the public understand the written description. These illustrations should use reference numbers or letters to point to specific components described in the text, but they must not contain descriptive text themselves [47].

3.4 Freedom to Operate (FTO)

Securing a patent does not automatically grant a researcher the right to sell or use their invention; they must first ensure they have Freedom to Operate (FTO). FTO means the legal ability to develop and market a product without infringing on the existing intellectual property rights of others. Even if an invention is a novel improvement, it might still incorporate a core technology that is currently patented by a competitor [48].

The FTO status is determined by conducting a clearance search. This process involves identifying live patents in the specific geographic regions where the researcher intends to operate and carefully analysing their claims. By comparing the researcher's product to the claims of these active patents, they can assess the risk of legal action and determine if they need to seek a license, redesign the product, or wait for existing patents to expire.

FTO analysis is essential for assessing business risks before commercialization. A researcher might find that their invention sits entirely within another company's patent area, or there might be an overlap that requires negotiation. Understanding the "IP position" early, including whether a competitor's patent is still in force, prevents costly legal battles and informs the strategy for moving the invention from the lab to the market.

3.5 Search and Documentation

Effective patenting begins with a thorough search of existing databases to understand the "prior art". Researchers should use keywords related to the problem, structure, and function of their invention. For a digital stethoscope, for example, keywords might include "handling noise," "accelerometer," and "adaptive noise cancellation". Utilizing Boolean search techniques (using AND, OR, NOT) and citation searches, looking both backward at older references and forward at newer ones, helps ensure no relevant documents are missed [49].

It is highly recommended to maintain a systematic way of documenting findings, such as a patent search tracking spreadsheet. This documentation should include notes on the objective assessment of other patents' claims and how they relate to the researcher's own work. By organizing this data, the researcher can articulate exactly how their invention differs from existing solutions, which is a requirement for the patent description.

Several resources are available for this research, including the PRH, the EPO, and Google Patents. These databases provide access to millions of patent documents, allowing researchers to see what has already been tried and where the "gaps" in current technology lie. A researcher should spend significant time reviewing a small subset of the most relevant patents in depth rather than skimming thousands of unrelated entries.

3.6 Timeline of the Process

The patent process follows a specific chronological path, starting from the initial conception and a patentability search. Once the researcher is confident in the novelty of the idea, a Provisional Application is often filed. This "filing date" is crucial because it establishes the "priority date," which is used to determine what counts as "state of the art" for all future examinations of that invention [49].

Within 12 months of the first filing, the researcher must decide whether to expand protection internationally. This 12-month window, known as the "priority year," allows the applicant to file a PCT (Patent Cooperation Treaty) application or national applications in other countries while still benefiting from the original priority date. During this time, the application is "patent pending," and the researcher can seek funding or partners while the technical examination proceeds.

The final stages involve prosecution and the national phase, typically occurring around 30 months after the initial filing. In this stage, individual patent offices like the PRH or the EPO conduct a detailed examination of the claims.

If the examiner is satisfied that the criteria for novelty and inventive step are met, the patent is granted. From start to finish, the process requires careful attention to deadlines to ensure that international rights are not lost.

3.7 Summary of Patentability

A patent search was conducted in the “Google Patents” database to evaluate the novelty and uniqueness of the proposed “Modular Smart Offloading Insole.” The search identified 15 related patents associated with wearable health technologies, smart footwear systems, gait monitoring, and pressure management devices. Among these, only four patents demonstrated partial similarities in terms of diabetic foot monitoring, ulcer prevention, or adaptive offloading mechanisms, as summarized in Table 4. Nevertheless, detailed technical analysis revealed significant differences between the proposed system and the existing patents.

Most of the reviewed inventions focused either on general physiological monitoring, motion tracking, or conventional orthotic structures without integrating a modular offloading architecture combined with intelligent sensing capabilities.

In contrast, the proposed device introduces a unique combination of customizable pressure redistribution modules, smart monitoring functions, and adaptable structural design specifically intended for diabetic foot management. Furthermore, none of the reviewed patents provided the same level of personalization, modularity, and real-time feedback integration envisioned in this study.

Table 4. Review of Existing Patents for the Modular Smart Offloading Insole

Publication Number	Patent Title
US-10729336-B1	Smart watch
US-11389652-B2	Monitoring and management of physiologic parameters of a subject
CN-110337265-B	Methods and devices for improving peripheral nerve function
US-10863084-B2	Video smart phone
US-11278765-B2	Methods and apparatus for power expenditure and technique determination during bipedal motion
EP-3370606-A1	Footwear system for ulcer or pre-ulcer detection [50]
US-2023281346-A1	Customized Foot Support and Systems and Methods for Providing Same [51]
US-2022133519-A1	Differential and variable stiffness orthosis design with adjustment methods, monitoring and intelligence [52]
US-2025213186-A	Expenditure to overcome air resistance during bipedal motion
US-2022117828-A1	Smart walking foot assembly with dynamic feedback [53]
US-2004196265-A1	System and method for finger held hardware device
US-2025295916-A1	Monitoring and management of physiologic parameters of a subject
US-2024315390-A1	Instrumented artificial intelligence (ai) driven motion tracking and alignment systems and methods for various applications
US-2023390094-A1	Differential and variable stiffness orthosis design with adjustment methods, monitoring and intelligence
WO-2024025998-A1	Custom exoskeleton configuration via data points

Elgendi et al. present a footwear-integrated system for detecting ulcer or pre-ulcer conditions, in which an array of temperature sensors is used to measure

thermal patterns across the plantar surface of the foot. The system applies pattern-recognition techniques to identify abnormal conditions that may lead to ulcer formation. However, because the sensing components are embedded within the shoe structure itself, this approach falls outside the scope of the present study, which focuses on insole-based solutions that can be used independently within different types of footwear [50].

Armstrong et al. describe a method for creating customized foot supports based on 3D scanning and plantar pressure or load distribution data. This approach enables the design and manufacturing of personalized insoles tailored to the user's foot geometry and biomechanics. Nevertheless, the resulting product is a passive orthotic device without embedded sensors or real-time monitoring capabilities, and therefore lacks smart functionality [51].

Smith et al. introduce a differential and variable stiffness orthosis designed for the ankle-foot complex, incorporating adjustable mechanical elements such as springs and joint mechanisms. This device aims to support movement, stabilize joints, and assist rehabilitation through controlled biomechanical modulation. Unlike smart insoles, which primarily focus on pressure monitoring and ulcer prevention, this system operates at a higher structural level and integrates mechanical actuation with intelligent control [52].

The patent US-2022117828-A1, titled Smart Walking Foot Assembly with Dynamic Feedback [53], proposes a rehabilitation-oriented system designed primarily for patients using assistive devices such as walkers or canes. The system measures load during walking and provides dynamic feedback to support proper weight-bearing and reduce excessive pressure on the affected limb. Unlike rehabilitation-focused external support systems, smart insoles are intended for continuous use inside footwear and primarily focus on plantar pressure monitoring, gait analysis, and the prevention of complications such as diabetic foot ulcers [53].

Researcher has filed a patent application (Application No. 20265267) with the PRH for an invention titled “Modular Sensing Offloading Insole System,” which is currently under examination.

4 Research Ethics and Research Permit Process

This section describes the research permit requirements, ethical considerations, and documentation preparation processes related to this study. It also outlines the application procedures followed at Metropolia University of Applied Sciences and the City of Helsinki to ensure that the research was conducted in accordance with institutional policies, data protection regulations, and the General Data Protection Regulation (GDPR).

4.1 Research Permission Requirements

Research conducted in universities or public organizations may require obtaining an official research permit. This permit ensures that the research is carried out in accordance with organizational policies, ethical standards, and data protection regulations [54].

Considering the required research field related to orthoses and wearable medical devices (smart insoles), three institutions can be mentioned:

- Metropolia University of Applied Sciences (Metropolia)
- City of Helsinki
- HUS (Helsinki University Hospital)

Table 5. Comparison of Research Permit Policies in Metropolia, HUS, and the City of Helsinki [54],[55],[56].

	Metropolia University of Applied Sciences	**City of Helsinki**	**HUS – Helsinki University Hospital**
When a research permit is required	When collecting data from Metropolia students or staff	When collecting data from social and health services staff or clients	When research involves HUS patients, staff, facilities, or registers
Typical research context	Student theses involving university participants	Research related to municipal social and health services	Clinical or healthcare research
Key prerequisite	Supervisor must approve thesis plan	Clarify research feasibility with the relevant unit	Must have a contact person or supervisor within HUS
Mandatory attachments	Research plan, consent form, privacy notice, participant information	Research plan, consent form, privacy notice, research materials	Research plan, consent form, privacy statement, confidentiality agreement
Ethical review requirement	Required if the study involves specific ethical risks	May be required depending on research design	Mandatory for research involving patients
Data protection compliance	Must follow GDPR and institutional data protection policies	Must comply with GDPR and municipal data protection rules	Strict compliance with GDPR and healthcare data regulations
Processing time	Approximately 2–3 weeks	Approximately 2 weeks – 1.5 months	Around 1 month for staff research, 3–4 months for patient research
After research completion	Thesis published in Theseus database	Research results should be reported to the city	Research completion must be reported through Researcher's Desktop

The selection of institutions depends on the target research population (e.g., professors, staff members, physicians, nurses, patients, etc.).

For example, in this research, because there was no need for clinical testing or for collecting data from patients, no request was submitted to HUS. It should also be noted that conducting research involving patients requires not only a research permit but also approval from an ethics committee.

The ethics committee approval process may take longer than the processes required by other institutions. Therefore, it is essential for the researcher to clearly define their timeline and research objectives when selecting the appropriate institution.

4.2 Preparation of Research Documentation

Before submitting the research permit application, all required research documentation and supporting materials were prepared. These documents were reviewed and approved by the thesis supervisor to ensure that the study followed institutional research ethics and data protection requirements.

The following documents were prepared prior to submitting the research permit application [54]:

- Research Plan
- Privacy Notice
- Informed Consent Form
- Information Sheet
- Data Management Plan

These documents provided detailed information about the research objectives, methodology, participant recruitment procedures, and data protection measures in accordance with applicable research ethics guidelines.

A Research Plan is a fundamental document that outlines the objectives, methodology, and implementation strategy of the study. It helps ensure that the research is scientifically sound, ethically conducted, and feasible within the available timeframe and resources (see Appendix 1).
When preparing a research plan, the researcher should clearly describe the following elements [54]:

- Research background and problem statement, explanation of the research topic and the problem the study aims to address.
- Research objectives and research questions, the main goals of the study and the specific questions the research intends to answer.
- Research methodology, the selected research approach, methods for data collection (such as interviews, surveys, or observations),
- Target participants or study population, description of the participants involved in the study and the criteria used for their selection.
- Data analysis approach, explanation of how the collected data will be analysed and interpreted.
- Ethical considerations, description of how participants' rights, privacy, and confidentiality will be protected.
- Data management considerations, information about how data will be collected, stored, protected, and eventually deleted.
- Research timeline, an estimated schedule for the different phases of the research process.

A well-structured research plan helps both the researcher and the reviewing institution evaluate whether the proposed study is methodologically appropriate, ethically acceptable, and practically achievable.

The Privacy Notice explains how participants' personal data will be processed during the research in accordance with the requirements of the GDPR. This document must clearly describe what type of data will be collected, the purpose of collecting the data, the legal basis for processing the data (typically participant consent in thesis research), and how the data will be stored and protected. It should also specify who will have access to the data, how long the data will be retained, and how it will be securely deleted after the completion of the research. In addition, the document must inform participants about their rights, such as the right to withdraw consent or request deletion of their data (see Appendix 2).

The Informed Consent Form is a document used to obtain voluntary permission from participants before they take part in the research. This form confirms that participants have received sufficient information about the study and understand

the purpose, procedures, and potential risks or benefits of participating. It must clearly state that participation is voluntary and that participants have the right to withdraw from the study at any time without negative consequences. The form usually includes confirmation of consent for specific activities, such as audio recording of interviews, and requires the participant's signature and date as evidence of informed agreement (see Appendix 3).

The Participant Information Sheet provides detailed information about the research project in a clear and understandable manner before participants decide whether to participate. This document typically explains the purpose of the study, the background of the research, the role of participants, and what participation involves (for example interviews, surveys, or testing a prototype). It also describes how confidentiality will be maintained and how the collected data will be used in the research results. The information sheet helps participants make an informed decision about their participation and ensures transparency in the research process (see Appendix 4).

To prepare the Data Management Plan, the DMPTuuli online tool was used. This tool is commonly recommended by Metropolia for planning how research data will be handled during a thesis project. In this system, a new plan was created by selecting Metropolia as the organization and choosing the thesis template. The plan was then completed by describing what type of data would be collected in the study, how the data would be stored securely during the research, who would have access to the data, and how the data would be deleted after the thesis was completed. Using this tool helps researchers ensure that research data is handled responsibly and in accordance with data protection regulations such as the General Data Protection Regulation (see Appendix 5).

4.3 Research Permit Application Process at Metropolia

After preparing the necessary research documentation, the research permit application was submitted to through the university's electronic application system and the application was completed using the official online form [54].

The research permit application process included the following steps:

1. Submission of the online research permit application form together with the required attachments.
2. Confirmation of application receipt, which was sent via email by the research permit review committee of Metropolia.
3. Review of the application by the Metropolia, during which the applicant may be asked to clarify or revise certain documents or provide additional information.
4. Revision and resubmission of the requested documents, if required by the review committee.
5. Final approval of the research permit, which was communicated to the researcher via email.

Data collection for the study was initiated only after the research permit had been officially granted.

4.4 Research Permit Application Process at City of Helsinki

After preparing the required research documentation, the research permit application was submitted to the City of Helsinki through its official electronic application. The application was completed using the designated online form [56].

The research permit process at the City of Helsinki included the following steps:

1. Submission of the online research permit application form together with all required supporting documents and attachments.
2. Confirmation of receipt of the application, which was sent via email by the responsible authority within the City of Helsinki.
3. Administrative and ethical review of the application, during which the applicant may be requested to clarify, modify, or supplement submitted materials.
4. Revision and resubmission of documents, if required based on feedback from the review process.

5. Final decision and approval of the research permit, which was officially communicated to the applicant via email.

4.5 Conclusion

To obtain a research permit in Finland, the researcher must first prepare all required documentation, including a research plan, privacy notice, informed consent form, participant information sheet, and data management plan. These documents must clearly describe the study objectives, methodology, ethical considerations, and data protection measures in compliance with GDPR.

Once the documentation is completed and approved by the thesis supervisor, the application is submitted through the official online system of the relevant institution, either Metropolia or the City of Helsinki. After submission, the process follows a standard sequence in both organizations: confirmation of receipt, administrative and ethical review, possible requests for clarification or revisions, resubmission if needed, and final approval communicated via email.

At Metropolia, the process is primarily academic and typically applies to studies involving students or staff, with a relatively faster review time. In contrast, the City of Helsinki process is more administrative and service-oriented, focusing on research involving municipal employees or services, with a stronger emphasis on ethical and data protection evaluation.

In both cases, data collection is only allowed after official approval of the research permit has been granted. The researcher has successfully obtained research permits from both Metropolia and the City of Helsinki.

5 Methods and Materials

This study adopts a qualitative research approach combined with the Conceptual Research Method (CRM) within a Design Science Research (DSR) framework, in order to systematically develop an innovative solution in the healthcare domain.

While layout technological know-how presents a preferred framework for growing and comparing an artifact, CRM become selected due to the fact it's specially designed to convert qualitative insights into conceptual layout knowledge. This study builds on hooked up DSR frameworks that emphasize iterative development, relevance, and rigor [57].

Accordingly, this study seeks to answer the following research question: How can a smart insole with pressure offloading capabilities be designed to support the prevention of diabetic foot ulcers?

The main objective of this study is to develop a new modular smart offloading insole (MSOLE) by systematically extracting and introducing expert knowledge into the design concept. In this context, CRM is used as a core method to bridge qualitative data and product innovation. The process begins by gathering expert insights through interviews, followed by identifying key concepts, patterns, and relationships within the data. These extracted elements are structured into design requirements and translated into functional and modular functionality of the MSOLE system.

The design and construction phase follows an iterative approach. First, raw interview data is analysed using coding techniques to identify keywords, recurring themes, and important concepts. These elements are grouped into high-level conceptual categories such as capture, offloading, modularity, usability, and personalization. Based on these conceptual clusters, design decisions are made that shape the architecture of the MSOLE, including modular sections (heel, metatarsal, and forefoot), sensor integration, and adaptive off-road capabilities. This approach ensures that the final concepts are directly based on expert knowledge rather than purely theoretical assumptions.

Solution proposals are qualitatively evaluated through expert validation. Participants evaluate the conceptual design, comment on its feasibility and usability, and suggest improvements. This method of evaluation is consistent with the nature of the product's early stages, a stage where conceptual validity and

practical relevance are far more important than quantitative performance measures.

CRM was chosen because it allows us to transform qualitative insights into structured design knowledge, making it particularly suitable for innovation-based research. Unlike purely descriptive qualitative methods, CRM supports abstraction, synthesis, and conceptual modelling that are essential to the development of new products like MSOLE. Furthermore, the combination of CRM and DSR ensures that research produces not only knowledge but also concrete, practical solutions.

Various strategies are used to ensure the accuracy of research. This includes systematic coding and classification of data, clear documentation of the concept development process, and traceability between interview results and design results. Semi-structured interviews allow for greater depth and flexibility, while verbatim transcription ensures accuracy. Ethical standards such as informed consent and GDPR compliance will be strictly adhered to throughout the study.

5.1 Participants and Sampling

Participants were selected using purposive sampling to ensure relevance and expertise. In total, five specialists took part in the study, recruited from Metropolia and VTT Technical Research Centre of Finland. The participants included professors, university staff, and industry experts with knowledge of smart insoles, wearable medical devices, and orthotic systems.

The inclusion criteria required participants to have professional or academic experience related to the design of medical insoles, wearable sensor technologies, or orthotics. Their interdisciplinary backgrounds allowed the study to incorporate both clinical and technical perspectives, which were essential for developing a comprehensive and practical solution.

Although the sample size was relatively small, efforts were made to gain deep and meaningful insights by asking focused and specialized questions. The emphasis was on the quality and relevance of the data rather than the quantity, which is consistent with qualitative and conceptual research approaches.

5.2 Data Collection and Analysis

Data were collected through semi-structured interviews conducted both online (via Zoom and Microsoft Teams) and in person. Each interview lasted approximately 60 to 90 minutes. An interview guide with open-ended questions was used to explore participants' experiences, expectations, and opinions related to clinical insoles and smart wearable systems.

All interviews were audio-recorded with participants' consent and transcribed verbatim. The analysis followed a CRM-based approach, starting with open coding to identify key terms and important statements. These were then grouped into categories and higher-level concepts through axial and selective coding.

The resulting conceptual structure formed the basis for the MSOLE design. Key themes such as pressure monitoring, adaptive offloading, modularity, user comfort, and real-time feedback were translated into specific design features. This approach ensured a clear and traceable link between raw data, conceptual insights, and the final product design.

6 Artifact Description

This section presents the MSOLE artifact by moving step by step from concept to implementation. It begins with an overview of the solution and the clinical problem it addresses, then explains how the design was developed using a structured CRM-based process grounded in expert input. The section continues by outlining the core design principles and system architecture, followed by the technical implementation and working logic of the system. Finally, it describes the

physical prototype, showing how all conceptual and technical elements come together in a coherent, functional device.

This study introduces and advances a novel healthcare solution known as the MSOLE. The system is designed to help prevent DFUs by integrating real-time plantar pressure sensing with manually offloading in a modular format. It directly addresses major shortcomings of conventional insoles, particularly their limited ability to personalize treatment, monitor pressure effectively, and respond dynamically to changing load conditions.

By bringing sensing and intervention together in one device, MSOLE supports continuous monitoring and immediate response to harmful pressure levels. This reduces prolonged exposure to high-risk areas under the foot and enables more proactive care. Its modular structure further enhances flexibility, allowing the insole to be tailored and adjusted according to individual patient needs and evolving clinical requirements (Figure 6).



Figure 6. Modular Smart Offloading Insole (MSOLE)

Positioned at the intersection of wearable medical technology, orthotic design, and intelligent sensing systems, MSOLE provides a comprehensive approach to foot health management. It continuously tracks pressure distribution and delivers instant feedback to the user through vibration or alert mechanisms when critical

thresholds are reached. At the same time, the modular design allows specific regions of the insole to be modified or replaced, giving both clinicians and users greater control over treatment and improving overall usability.

6.1 CRM-Based Design Process and Concept Development

The method started with familiarization, wherein all interview recordings have been transcribed and thoroughly reviewed more than one times. This step enabled a deep expertise of members' views on clinical insoles, wearable technologies, and DFU management.

Next part, open coding and key-word extraction was conducted. Key phrases regularly cited via way of means of members blanketed strain distribution, comfort, customization, sensor accuracy, real-time feedback, and modularity. These key phrases meditated each medical issues and technical requirements and in the concept development phase, these keywords were transformed into higher-level design concepts. For example, "pressure distribution" and "sensor accuracy" were translated into the concept of a sensing system, while "comfort" and "fit" contributed to ergonomic design. Similarly, "customization" and "modularity" led to the concept of interchangeable components (Figure 7).

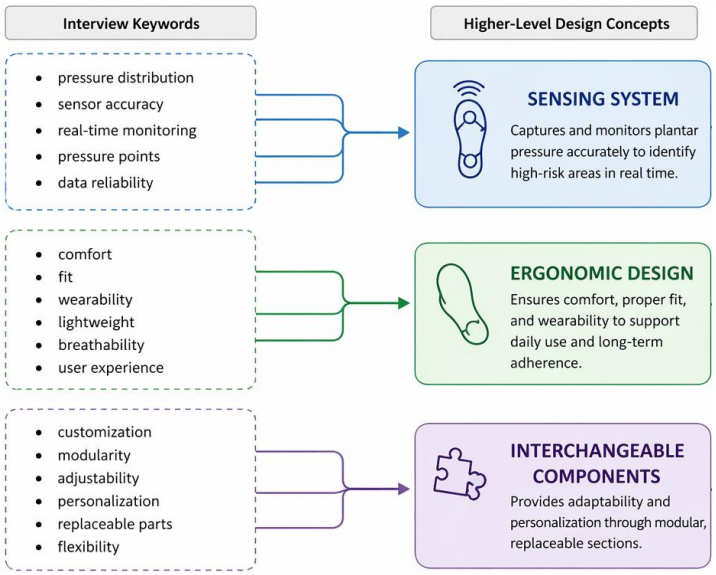


Figure 7. Conceptual Mapping of Interview Keywords to Higher-Level Design Concepts

CRM was employed to systematically refine and advance the design, ensuring that the final artifact is directly grounded in expert knowledge. Data were collected through semi-structured interviews with five specialists from Metropolia and VTT Technical Research Centre of Finland, forming the foundation of the design process.

These concepts were then organized through axial coding into major thematic groups, including “sensing and monitoring,” “Offloading and pressure redistribution,” “modularity and adaptability,” and “user experience and comfort.” Subsequently, a conceptual mapping process was conducted to identify the relationships among these themes. For example, sensing enables pressure detection; this detection activates feedback mechanisms, which in turn guide adjustments related to pressure reduction, and this logical flow collectively forms the functional foundation of the MSOLE system (Figure 8).

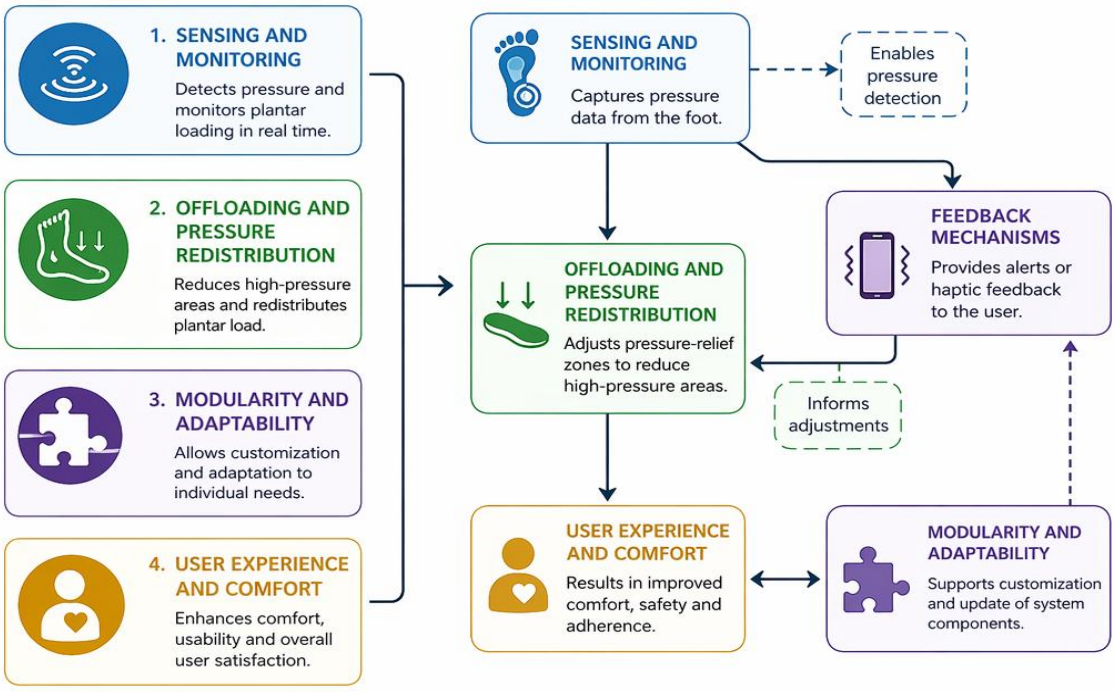


Figure 8. Axial Coding Themes and Their Functional Interconnections in MSOLE

In the final stage, the transformation of concepts into design requirements, each concept was directly linked to a specific feature of the artifact. For example, sensing is translated into embedded pressure sensors, feedback into alarm or

vibration systems, modularity into replaceable insole components, and load relief into adjustable pressure distribution areas (Figure 9). This structured process enabled clear and continuous traceability from raw interview data to final design decisions, greatly enhancing the effectiveness and relevance of the developed solution.

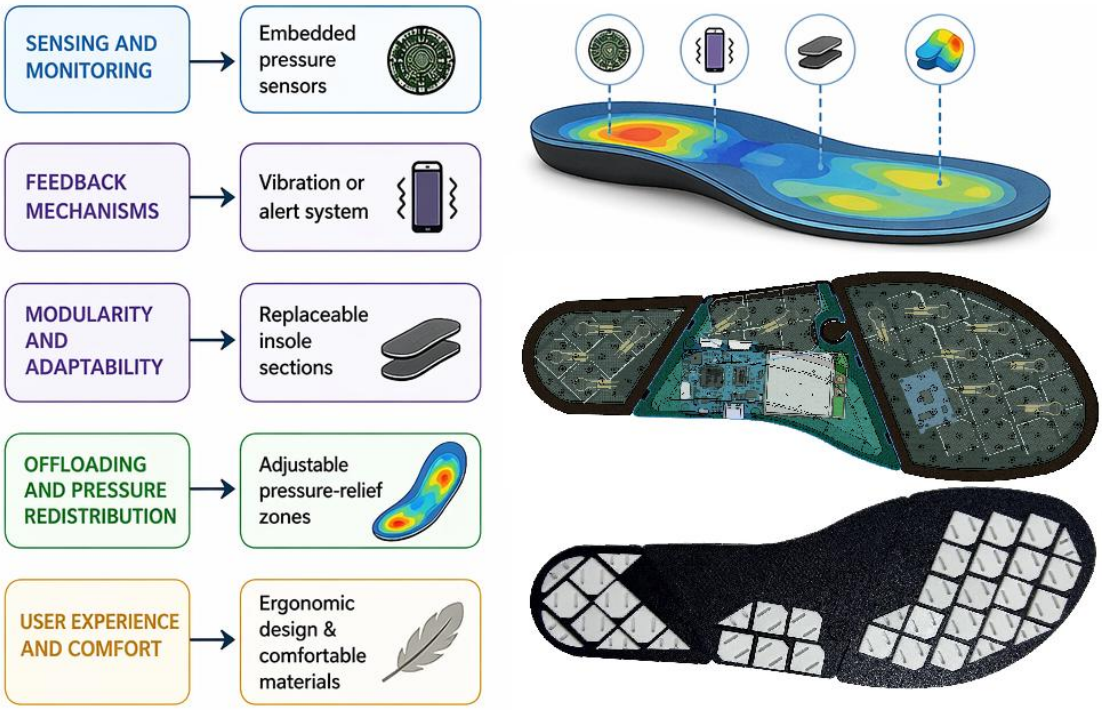


Figure 9. Translation of Conceptual Themes into Design Requirements for the MSOLE System

6.2 Design Principles

The development of the MSOLE is guided by several key design principles derived from both literature and expert input:

- Personalization: The system must adapt to individual patient needs.
- Modularity: Components should be replaceable and configurable.
- Real-time responsiveness: Immediate detection and feedback are essential.
- Usability and comfort: The insole must be comfortable for daily use.
- Clinical relevance: The design should support practical medical applications.

6.3 System Architecture

The MSOLE is built using a modular architecture, meaning it consists of multiple detachable sections. These sections typically correspond to the forefoot, midfoot, and rearfoot regions of the foot. Each module can function independently while still contributing to the overall system performance. This segmented design provides flexibility in design and functionality. It also allows for targeted pressure management in different anatomical areas (Figure 10).

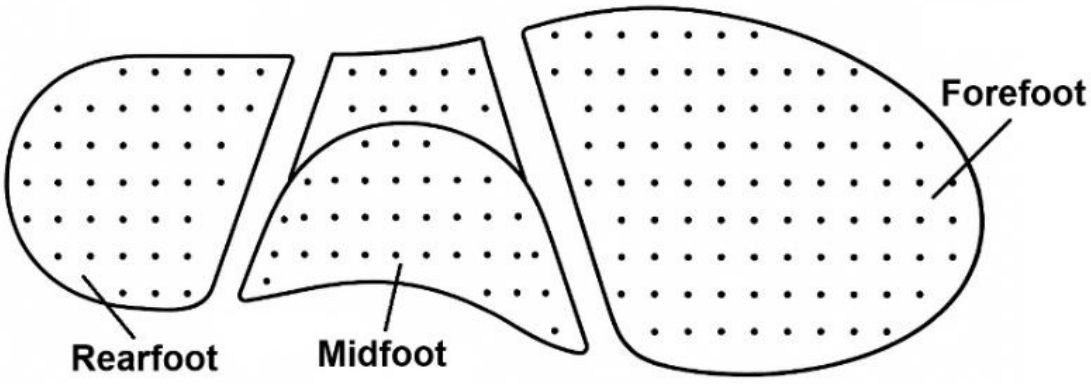


Figure 10. Modular structure of the MSOLE showing forefoot, midfoot, and rearfoot sections

Modular design allows each part of the sole to be optimized for a specific function. For example, the forefoot module is designed to control propulsion during walking. Midfoot module supports stability and weight distribution. The rearfoot module focuses on shock absorption during heel strike. This division improves the overall efficiency of the system. This ensures that all areas of the foot are properly supported.

The top layer of the insole is designed to be in direct contact with the user's foot. This layer is made of soft and comfortable material, which improves the user experience. This layer distributes initial pressure and reduces discomfort during long-term use. It also contributes to breathability and hygiene. Overall, it plays a significant role in user comfort and portability.

The sensor layer is integrated into the structure below the cover layer. This layer contains pressure sensors distributed throughout the insole. These sensors

continuously monitor how pressure is applied to various parts of the foot. The collected data provides a real-time map of foot pressure distribution. This information is essential for detecting high pressure areas.

6.4 Technical Implementation

The MSOLE system's algorithm begins by continuously collecting plantar pressure and motion data via a pressure sensor and inertial measurement unit (IMU) embedded in the insole. This raw data is sent to a processing unit where initial pressure and motion analysis is performed to identify patterns and abnormal load distributions.

The system then repeats the measurement and analysis process to ensure accurate and real-time detection of critical pressure points. Based on the analyzed data, the high-pressure areas are identified. In the next step, the clinician manually performs offloading according to these pressure zones. This is done by removing or keeping specific blocks under the insole to create empty spaces in targeted areas, which helps reduce pressure. The patient then wears the shoe again, and another pressure measurement is performed to achieve the lowest possible pressure in the affected areas (Figure 11).

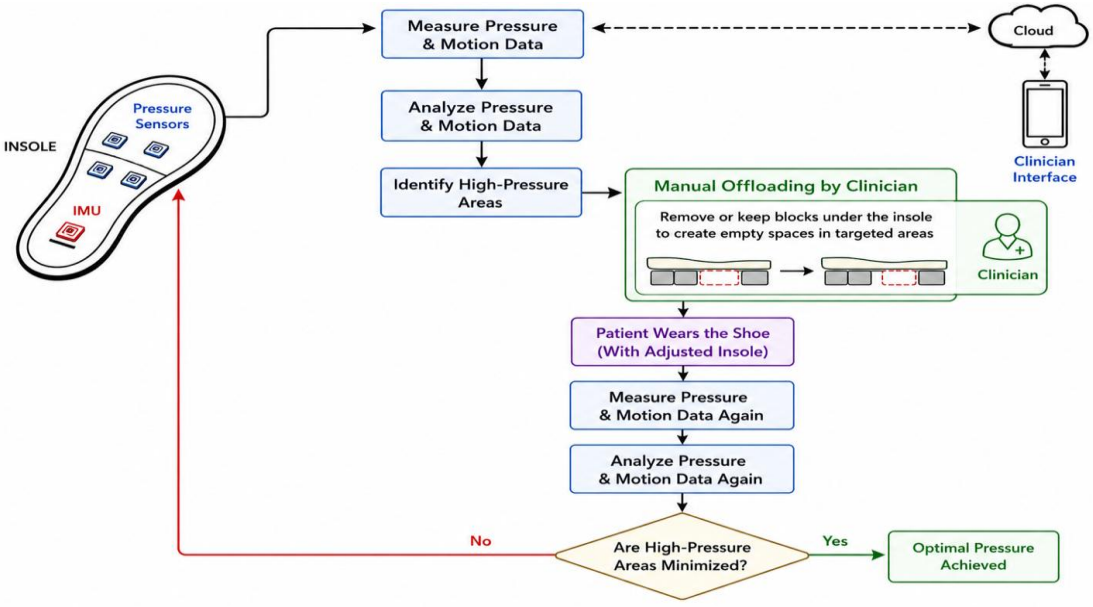


Figure 11. Flowchart of the MSOLE system algorithm for real-time pressure monitoring and adaptive control

The advantage of this system is its compact and lightweight design. Unlike pneumatic or mechanical systems, it does not require large components. This makes daily use more comfortable. It also improves compatibility with standard shoes. Users can wear it without major modifications.

The modular design also simplifies maintenance and replacement. If a module is damaged, it can be replaced individually. There is no need to replace the entire in-sole. This reduces costs and extends the life of the product. It also makes future up-grades easy.

Another advantage is the possibility of customizing the system. The different modules can be configured according to the user’s needs. For example, users with heel spurs may need different settings. This flexibility increases personalization. This ensures better results for different users.

The system is designed to be user-friendly and unobtrusive. It does not affect the user's natural walking behaviour and physical function. Users can move normally while the system is running in the background. This makes it suitable for long-term use. Comfort and ease of use are of paramount importance.

6.5 Prototype Design Description

The MSOLE prototype reflects the integration of all conceptual and technical elements derived from the CRM analysis. The insole is physically divided into three main areas, and each area contains several interchangeable modules with different properties.

The prototype includes embedded sensors in the main pressure zones, a compact electronic unit for processing, and a feedback mechanism to alert the user. Modular sections can be removed and replaced with alternative components, allowing for customization to suit the user's condition and treatment plan.

This design directly reflects insights from expert interviews, with a consistent focus on flexibility, accuracy, and ease of use. As a result, the MSOLE prototype represents a data-driven, user-informed innovation that bridges the gap between clinical needs and wearable technology solutions.

6.6 Evaluation

Given the design-focused nature of this study, the evaluation was based on qualitative validation, expert feedback, and early prototype feasibility rather than large-scale clinical testing.

The evaluation criteria were defined based on the research objectives and included: (1) sensing and monitoring, (2) offloading and pressure redistribution, (3) modularity and adaptability, and (4) user experience and comfort. These criteria were derived from the literature and expert interviews, ensuring alignment with real-world needs.

The testing methods included both functional and conceptual evaluation of the prototype. Functional testing focused on whether the sensors could detect pressure points and whether the feedback system (e.g., vibration alerts) responded correctly. Usability testing was carried out through expert evaluation, where participants assessed the design, structure, and interaction of the system. Since the project is still in an early stage, performance testing remained preliminary and focused on feasibility rather than detailed quantitative results.

User evaluation was conducted indirectly through expert interviews with specialists from Metropolia University of Applied Sciences and VTT Technical Research Centre of Finland. Participants shared insights based on their experience with orthotics, wearable medical devices, and smart insole technologies. Their feedback highlighted the importance of comfort, personalization, and reliable sensing, and confirmed that the MSOLE concept addresses key user needs.

Technical validation showed that the proposed system is feasible from an engineering perspective. The integration of pressure sensors, a microcontroller, and feedback mechanisms was successfully demonstrated at the concept and prototype stages. Although some challenges remain, such as optimizing sensor placement and improving energy efficiency, the overall system design is practical and coherent.

The qualitative evaluation indicates that the MSOLE design meets the key requirements of a wearable healthcare device, including modularity, adaptability, and a user-cantered approach. The structured development process, based on CRM and DSR, further supports the reliability and traceability of the solution.

The results also suggest that MSOLE has strong potential to address the main challenge of DFU prevention. By combining real-time pressure monitoring with adaptive offloading, the system enables both preventive and responsive interventions that can help reduce the risk of ulcer development.

7 Discussion and Conclusions

This study demonstrates that it is feasible to design a smart insole that both monitors plantar pressure and actively offloads it to help prevent diabetic foot ulcers. By combining sensing, real-time feedback, and a modular architecture in a single system, the MSOLE concept addresses a complex and clinically important problem that has typically been approached in fragmented ways.

A review of existing solutions shows a clear divide in the field. Commercial products such as orpyx, moticon, and Sensoria focus on pressure monitoring without actively reducing harmful loads. In contrast, traditional orthotics and devices from companies like darco provide pressure relief but lack sensing and feedback capabilities. MSOLE was intentionally designed to bridge this gap by integrating both functions into a unified, adaptive system, while also introducing a level of personalization that is largely missing in current technologies.

The project achieved its main objectives by delivering a functional prototype and validating the MSOLE concept. Expert knowledge from clinical, engineering, and podiatric domains was systematically translated into design features through a structured research process. The resulting system integrates pressure sensing, adaptive manually offloading using a light-activated responsive material layer, modular segmentation (forefoot, midfoot, rearfoot), and real-time feedback.

Methodologically, the study shows how CRM can be effectively embedded within a DSR framework to move from qualitative insight to a tangible artifact. Interview data were analysed, translated into design requirements, and then implemented as concrete technical features. This structured and traceable pathway from knowledge to prototype contributes to the broader literature on healthcare design and wearable medical devices and also produced a conference paper and a patent application filed with the the Finnish Patent and Registration Office (Application No. 20265267), reflecting both its academic relevance and commercial potential.

However, several limitations remain. Battery life, electronic miniaturization, and scalable manufacturing processes require further development. In addition, the qualitative evaluation involved five expert participants, which is appropriate for early-stage design research focused on depth rather than statistical generalization. While their feedback was consistent and informative, it reflects expert judgment within a limited context. Broader studies involving patients and clinicians are needed to assess generalizability and real-world performance and future work should focus on clinical validation through controlled trials, quantitative performance testing, and continued refinement of the prototype. A key next step is a structured feasibility study with patients at high risk of diabetic foot ulcers to evaluate usability and collect real-world pressure data.

Overall, MSOLE points toward a shift in diabetic foot care, from reactive treatment after ulcers develop to proactive prevention. By continuously sensing and adapting to changes in plantar pressure, the system supports a more personalized and preventive approach to care, aligning with broader trends in modern healthcare.

References

1 Sharma S, Rai M, Mahantesh L, Belehalli P. Offloading in diabetic foot ulcers: a comprehensive review with focus on the Indian context. Discov Med. 2025; doi:10.1007/s44337-025-00285-x

2 Armstrong DG, Boulton AJM, Bus SA. Diabetic foot ulcers and their recurrence. N Engl J Med. 2017;376(24):2367-2375. doi:10.1056/NEJMra1615439

3 Schaper NC, van Netten JJ, Apelqvist J, Lipsky BA, Bakker K. Prevention and management of foot problems in diabetes: a summary guidance for daily practice 2015, based on the IWGDF guidance documents. Diabetes Metab Res Rev. 2015;32(Suppl 1):7-15. doi:10.1002/dmrr.2695

4 Schaper NC, van Netten JJ, Apelqvist J, Bus SA, Hinchliffe RJ, Lipsky BA. Practical guidelines on the prevention and management of diabetic foot disease (IWGDF 2019 update). Diabetes Metab Res Rev. 2020;36(S1):e3266. doi:10.1002/dmrr.3266

5 Korada H, Maiya A, Rao SK, Hande M. Effectiveness of customized insoles on maximum plantar pressure in diabetic foot syndrome: a systematic review. Diabetes Metab Syndr. 2020;14(5):1093-1099. doi:10.1016/j.dsx.2020.06.041

6 Ren Y, Wang H, Song X, Wu Y, Lyu Y, Zeng W. Advancements in diabetic foot insoles: a comprehensive review of design, manufacturing, and performance evaluation. Front Bioeng Biotechnol. 2024; 12:1394758. doi:10.3389/fbioe.2024.1394758

7 Hingorani A, LaMuraglia GM, Henke P, Meissner MH, Loretz L, Zinszer KM, et al.The management of diabetic foot: a clinical practice guideline by the Society for Vascular Surgery in collaboration with the American Podiatric Medical Association and the Society for Vascular Medicine. J Vasc Surg. 2016;63(2 Suppl):3S-21S. doi: 10.1016/j.jvs.2015.10.003

8 Collings R, Freeman J, Latour JM, Paton J. Footwear and insole design features for offloading the diabetic at risk foot: a systematic review and meta-analyses. Endocrinol Diabetes Metab. 2020;4(1): e00132. doi:10.1002/edm2.132

9 Ahmed S, Butterworth P, Barwick A, et al. Footwear and insole design parameters to prevent occurrence and recurrence of neuropathic plantar forefoot ulcers in patients with diabetes: a series of N-of-1 trial study protocol. Trials. 2022;23(1):1017. doi:10.1186/s13063-022-06968-5

10 Jarl G, Rusaw DF, Terrill AJ, Barnett CT, Woodruff MA, Lazzarini PA. Personalized offloading treatments for healing plantar diabetic foot ulcers. J Diabetes Sci Technol. 2022;17(1):45-55. doi:10.1177/19322968221101632

11 Lockhart M, Dinneen SF, O'Keeffe DT. Plantar pressure measurement in diabetic foot disease: a scoping review. J Diabetes Investig. 2024;15(8):990-999. doi:10.1111/jdi.14215

12 Castro-Martins P, Marques A, Coelho L, Vaz M, Baptista JS. In-shoe plantar pressure measurement technologies for the diabetic foot: a systematic review. Heliyon. 2024;10(9): e29672. doi: 10.1016/j.heliyon. 2024.e29672

13 Bardill JR, Laughter MR, Stager M, Liechty KW, Krebs MD, Zgheib C. Topical gel-based biomaterials for the treatment of diabetic foot ulcers. Acta Biomater. 2022; 138:73-91. doi: 10.1016/j.actbio.2021.10.045

14 Li B, Lin A, Huang J, Xie J, Liu Q, Yang C, Zhang Z. Total contact casts versus removable offloading interventions for the treatment of diabetic foot ulcers: a systematic review and meta-analysis. Front Endocrinol. 2023;14. doi:10.3389/fendo.2023.1234761

15 Shi QQ, Li PL, Yick KL, Li NW, Jiao J. Effects of contoured insoles with different materials on plantar pressure offloading in diabetic elderly during gait. Sci Rep. 2022; 12:15395. doi:10.1038/s41598-022-19814-0

16 Bus SA, Waaijman R, Arts M, de Haart M, Busch-Westbroek T, van Baal J, Nollet F. Effect of custom-made footwear on foot ulcer recurrence in diabetes: a multicenter randomized controlled trial. *Journal of Biomechanics*. 2017; 59:63–70. doi: 10.1016/j.jbiomech.2017.06.028

17 Bus SA, Armstrong DG, van Deursen RW, Lewis JEA, Caravaggi CF, Cavanagh PR. IWGDF guidance on footwear and offloading interventions to prevent and heal foot ulcers in patients with diabetes. Diabetes/Metabolism Research and Reviews. 2016;32(S1):25–36 . doi:10.1002/dmrr.2697

18 Wang Q, Zou Y, Wang Y, Xiang C, Wang Z, Shi Q, et al. A wireless, self-powered smart insole for gait monitoring and recognition via nonlinear synergistic pressure sensing. *Science Advances*. 2025;11: eadu1598. doi:10.1126/sciadv. adu1598.

19 Gurney JK, Kersting UG, Rosenbaum D. Between-day repeatability of pedobarographic measurements in healthy adults. Gait Posture. 2008;28(3):533-537. doi: 10.1016/j.gaitpost.2008.03.012

20 Minty E, Bray E, Bachus CB, Armstrong DG, Bus SA, Lázaro-Martínez JL, et al. Preventative sensor-based remote monitoring of the diabetic foot in clinical practice. *Sensors (Basel)*. 2023; 23(15):6712 . doi:10.3390/s23031100.

21 van Netten JJ, Raspovic A, Lavery LA, et al. Prevention of foot ulcers in the at-risk patient with diabetes: a systematic review. Diabetes Metab Res Rev. 2019;36(S1): e3270. doi:10.1002/dmrr.3270

22 International Diabetes Federation. IDF diabetes atlas. 10th ed. Brussels: International Diabetes Federation; 2021. Available from: https://www.diabetesatlas.org

23 American Diabetes Association Professional Practice Committee. Introduction and methodology: Standards of Care in Diabetes-2023. *Diabetes Care*. 2023;46(Suppl 1)-S4. doi:10.2337/dc23-SINT

24 National Institute for Health and Care Excellence. Diabetic foot problems: prevention and management (NICE guideline NG19, updated 2019). London: NICE; 2019. Available from: https://www.nice.org.uk/guidance/ng19

25 Armstrong DG, Tan TW, Boulton AJM, Bus SA. Diabetic foot ulcers: a review. JAMA. 2023;330(1):62-75. doi:10.1001/jama.2023.10578

26 Abbott CA, Chatwin KE, Foden P, et al. Innovative intelligent insole system reduces diabetic foot ulcer recurrence at plantar sites: a prospective, randomised, proof-of-concept study. Lancet Digit Health. 2019;1(6): e308-e318. doi:10.1016/S2589-7500(19)30128-1

27 Najafi B, Chalifoux CB, Everett JB, Razjouyan J, Brooks EA, Armstrong DG. Cost effectiveness of smart insoles in preventing ulcer recurrence for

people in diabetic foot remission. Wound Care Management. 2018;1(1):5-7.

28 Orpyx Medical Technologies. (2026). Orpyx® SI sensory insoles. https://www.orpyx.com

29 Moticon GmbH. (2026). Moticon® sensor insoles for gait and motion analysis. https://moticon.com

30 FeetMe SAS. (2026). FeetMe® smart insoles. https://www.feetme.fr

31 Sensoria Health Inc. (2026). Sensoria® wearable gait monitoring solutions. https://www.sensoriahealth.com

32 Novel GmbH. (2026). Pedar-X® in-shoe pressure measurement system. https://www.novel.de

33 Boogio Inc. (2026). Boogio smart wearable insoles. https://www.boogio.com

34 ARION. (2026). ARION smart running insoles. https://www.arion.run

35 Salted Venture. (2026). Salted smart insole technology. https://salted.ltd

36 TG0 Ltd. (2026). TG0 pressure sensing technology. https://tg0.co.uk

37 Nurvv Ltd. (2026). NURVV running smart insoles. https://www.nurvv.com

38 PediFix Inc. (2026). PediFix® diabetic foot care products. https://www.pedifix.com

39 DARCO International Inc. (2026). DARCO® offloading and diabetic foot solutions. https://www.darcointernational.com

40 Verified Market Research. *Smart insole market size, scope, growth, trends and forecast* [Internet]. Lewes (DE): Verified Market Research; 2025 [cited 2026 May 11]. Available from: Verified Market Research report

41 European Wound Management Association (EWMA). *Diabetic foot* [Internet]. Copenhagen: EWMA; 2024 [cited 2026 May 11]. Available from: EWMA Diabetic Foot Resources

42 Research and Markets. *Diabetic foot ulcer treatment market by ulcer type, treatment type, end user and geography – global forecast to 2028* [Internet]. Dublin: Research and Markets; 2024 [cited 2026 May 11]. Available from: Diabetic foot ulcer treatment market report

43 European Parliament and Council of the European Union. Regulation (EU) 2017/745 on medical devices (EU MDR), Annex VIII: Classification rules [Internet]. 2017 [cited 2026 May 11]. Available from: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32017R0745

44 U.S. Food and Drug Administration. De novo classification process (evaluation of automatic class III designation) [Internet]. 2023 [cited 2026 May 11]. Available from: https://www.fda.gov/medical-devices/premarket-submissions-selecting-and-preparing-correct-submission/de-novo-classification-request

45 Finnish Patent and Registration Office (PRH). Patents in Finland [Internet]. [cited 2026 May 11]. Available from: https://www.prh.fi/en/patents.html

46 Singer M, Stauder D, editors. *The European Patent Convention: a commentary*. 3rd English ed. London: Sweet & Maxwell; 2016.

47 Visser D. Visser's annotated European Patent Convention. Munich: European Patent Office; 2023.

48 Pagenberg J, Hacon R, editors. Concise European patent law. Alphen aan den Rijn: Kluwer Law International; 2020.

49 Veronese A, Watchorn D. Procedural law under the EPC 2000. Alphen aan den Rijn: Kluwer Law International; 2018.

50 Elgendi A, Jonkman M, de Vries J. Footwear system for ulcer or pre-ulcer detection. European Patent EP3370606A1. 2018.

51 Armstrong DB, Lavery LA, Frykberg RG. Customized foot support and systems and methods for providing same. U.S. Patent Application US2023/281346A1. 2023.

52 Lerner ZF, Tagoe E, Cuddeback C. Differential and variable stiffness orthosis design with adjustment methods, monitoring and intelligence. US Patent US2022133519A1; 2022.

53 Smart walking foot assembly with dynamic feedback. US Patent US2022117828A1; 2022.

54 Metropolia University of Applied Sciences. Research, development and innovation permit and ethical guidelines [Internet]. Espoo: Metropolia University of Applied Sciences; [cited 2026 May 11]. Available from: Metropolia research permits and ethical guidelines

55 City of Helsinki. Research permits in social and health services [Internet]. Helsinki: City of Helsinki; [cited 2026 May 11]. Available from: City of Helsinki research permits in social and health services

56 HUS – Helsinki University Hospital. Instructions for researchers and research permits [Internet]. Helsinki: HUS; [cited 2026 May 11]. Available from: HUS research permits and instructions for researchers

57 Peffers K, Tuunanen T, Rothenberger MA, Chatterjee S. A design science research methodology for information systems research. J Manag Inf Syst. 2008;24(3):45-77. doi:10.2753/MIS0742-1222240302

Appendix 1 - Research Plan

1. Background and Aims of the Research

Diabetic Foot Ulcers (DFUs) are a serious and common complication of diabetes, mainly caused by peripheral neuropathy and abnormal plantar pressure. Due to loss of protective sensation, patients are often unaware of excessive pressure or minor injuries, which can develop into chronic ulcers and, in severe cases, lead to infection and amputation. Pressure offloading is therefore a key preventive and therapeutic strategy.

Existing solutions mainly fall into two categories: pressure offloading insoles and smart monitoring insoles. While offloading insoles reduce pressure mechanically, they lack real-time feedback. Smart insoles can monitor plantar pressure but often do not actively reduce it. This gap limits their preventive effectiveness.
The aim of this research is to design and evaluate a Modular Smart Offloading Insole (MSOLE) that integrates both pressure monitoring and pressure offloading in a single personalized device. By using a modular structure, the insole can be adapted to individual foot anatomy, pressure distribution, and clinical needs, offering a more effective and user-cantered approach to DFU prevention.

2. Rationale for the Research

2.1 Link to Previous Research

Previous research has clearly demonstrated that high plantar pressure is a primary risk factor for the development and recurrence of diabetic foot ulcers. Clinical guidelines and systematic reviews emphasize pressure offloading as one of the most effective preventive measures. Studies on smart insoles have shown promising results in early detection of pressure abnormalities and reduction of ulcer recurrence through real-time feedback. At the same time, research on custom and multi-density insoles confirms their effectiveness in redistributing pressure and improving comfort.

However, most existing solutions focus either on monitoring or on mechanical offloading, rarely combining both in an integrated and adaptable way. Furthermore, traditional custom insoles are time-consuming and costly to produce, while smart insoles often suffer from durability and usability issues.

2.2 Theoretical and Methodological

The project is grounded in biomechanical theories of plantar pressure distribution and user-cantered medical device design. It assumes that personalized pressure offloading, combined with continuous monitoring, can significantly reduce DFU risk. Methodologically, the study follows an exploratory qualitative approach, which is suitable for investigating under-researched design concepts. Expert knowledge is treated as a critical source of insight for defining functional, technical, and usability requirements. The modular design concept is based on the premise that adaptability and replaceability can improve both clinical effectiveness and device lifespan.

3. Objectives of the Research and Scientific Impact

3.1 Objectives and Research Questions

The aim of this study is to enhance the efficiency and improve the design of smart insoles for the prevention and/or treatment of diabetic foot ulcers.

Key research questions include:
Can a modular smart insole effectively support pressure offloading for DFU prevention?
Which components should be modular, adjustable, or replaceable?
What technical and usability challenges arise in designing MSOLE?
How can modularity support personalization for different patients?

The main reason diabetic foot ulcers (DFUs) develop is repeated pressure or mechanical stress on areas of the foot that no longer feel pain due to nerve damage. Since people with sensory neuropathy cannot sense pain properly, these wounds often go unnoticed until they become serious. That's why one of the most important steps in early DFU care is "pressure offloading," which means using insoles or special devices to take pressure off the affected area.

Research shows that this really works. For instance, about 69% of studies report that insoles with an arch-shaped design increase the contact area of the foot and lower the pressure on vulnerable spots. In fact, arch supports alone reduced average plantar pressure by around 37.5 kPa (with 95% confidence, $p < 0.03$). By spreading the load more evenly, these designs reduce stress on high-risk areas, especially under the forefoot.

Other solutions like multi-density insoles, soft pads placed under the metatarsal region, and shoes with rocker soles have also been shown to cut down pressure and lower the chance of ulcers coming back . Taken together, the evidence suggests that the best results come from a combination of methods that are tailored to the patient. When biomechanics, personal needs, and new technology are considered together, DFU care becomes much more effective.

Personalized offloading is becoming especially important. Instead of depending on one standard device, the focus is shifting toward finding the right balance between reducing pressure, lowering shear forces, managing walking activity, and making sure patients actually use and accept the devices. Looking ahead, developments such as modular designs, advanced manufacturing techniques, and smart sensors will allow clinicians to adapt insoles and footwear to each patient's healing process .

Another review study mentioned that besides looking at the wider community, it is also important to focus on the personal needs of patients with diabetic foot ulcers. These needs can be met by using devices that are better in terms of cost, availability, and technical features .

3.2 Expected Results and Scientific Impact

The expected outcome is a clear design framework for a modular smart offloading insole, informed by expert perspectives. The novelty of the project lies in integrating modularity, pressure offloading, and smart monitoring into a single conceptual device. This approach contributes to current scientific discussions by addressing a known gap between monitoring technologies and mechanical offloading solutions. Beyond academia, the results may support future product development, inform orthotic education, and contribute to more cost-effective, patient-centered DFU prevention strategies in clinical practice.

4. Data Collection Plan

4.1 Target Population

The target population for obtaining appropriate answers in this study consists of university staff and faculty members, as well as medical specialists in the fields of diabetic foot ulcers and orthotics (medical insoles). Since the research is at an early stage and aims to develop a practical idea and an effective prototype, clinical testing is not required. Therefore, no research or data collection will be conducted with patients, and ethical approval from a medical committee is not required.

4.2 Sampling strategy

Considering that the statistical population atMetropolia University in the fields of orthotics, diabetes, and foot ulcers is limited, it can therefore be stated that three groups (faculty members, staff, and students with relevant work experience) will be selected for interviews. Each group will be categorized based on their knowledge and skills, and data will be collected from them.

4.3 Data sources & Pilot evaluation

Data will be collected through a qualitative study using interviews. The purpose of the interviews is to evaluate and try the functionality and features of the proposed device, MSOLE. Different scenarios, such as system start-up and pressure measurement, will be used to examine technical and design limitations. Since the project is at an early stage, expert and academic feedback is essential for improving the design. The results of the interviews and device evaluate are expected to contribute directly to the further development and refinement of the proposed design. The data collection process will begin with an explanation of how the device works. Participants will then be asked to set up and start the device themselves, after which the interview will begin. They will be asked to provide feedback on its usability and design structure. This will be followed by more technical and specialized questions. Finally, participants will be asked to share their overall opinions and suggestions to conclude the session.

4.4 Ethics & data protection

Written informed consent will be obtained from all participants. No personal information will be collected in the registration form; only participant codes (e.g., P1, P2…) will be used during analysis. Audio and textual data will be securely stored on the researcher’s personal laptop and internal hard drive. For backup purposes, the data will also be stored on a password-protected external hard drive.

5. Analysis Plan

5.1 Analytic framework & coding

The qualitative data collected through semi-structured interviews will be analyzed using thematic analysis. The interview recordings will be transcribed word for word and analyzed through a systematic coding process. First, open coding will be used to identify meaningful statements related to device performance,

usability, technical functionality, and design features of the MSOLE device. Next, during axial coding, these initial codes will be grouped into broader categories such as technical limitations, user experience, and clinical relevance. Finally, selective coding will be applied to develop the main themes that directly address the research question and inform further improvements in the design of the proposed smart insole.

5.2 Credibility & saturation

Several strategies will be used to strengthen the credibility of the qualitative findings. By collecting perspectives from different expert groups, including expert in clinicians, orthotists, and academic faculty members, perspective triangulation will be achieved, leading to a more comprehensive and balanced understanding of the research topic. In addition, the coding process and emerging themes may be reviewed by academic supervisors or peers to support peer review and reduce interpretive bias. Transparency will be maintained by documenting analytical decisions, code development, and revisions throughout the analysis process.

5.3 Reporting, quotes & anonymization

The results of the thematic analysis will be reported by presenting the identified themes and supporting them with direct quotations from interview participants. These quotations will be used to illustrate key findings related to the technical performance, design structure, and practical application of the MSOLE device. To ensure ethical research practices and protect participant confidentiality, all data will be anonymized, and no personally identifiable information will be disclosed. Participants will be referred to using coded identifiers based on their professional roles, such as clinicians, orthotists, or academic experts.

5.4 Tools and planned outputs

During this research, interviews will be conducted in two ways: in-person interviews and online interviews. Online interviews will be carried out using digital

platforms such as Zoom and Microsoft Teams, which allow flexible scheduling, audio and video recording, and effective communication with participants in different locations. All interviews, whether in person or online, will be recorded with the participants' consent and later transcribed word for word for analysis.

Microsoft Excel 365 will be used to store, organize, and analyze the collected data. It enables basic descriptive statistical analysis to summarize the data and identify general patterns and Power Query, integrated within Excel 365, will be used for data cleaning and transformation. It ensures a consistent, accurate, and repeatable data preparation process with automatic updates.

6. Risks & Limitations

One of the main risks is particimpant selection limition. This may occur because the collected opinions could come from a limited group of experts or specific academic professionals and may not fully represent all potential users. To reduce this risk, experts with different clinical and technical backgrounds can be included, and their perspectives can be compared. As a limitation, this study mainly focuses on the feasibility of the design rather than proven clinical effectiveness. Therefore, the findings cannot be fully generalized to diabetic patients and will require future clinical studies for validation.

When examining the possible limitations of MSOLE in relation to the question "What limitations might the MSOLE create for patients with diabetic foot ulcers?" participants may not fully express real concerns because the device is still a prototype. As a result, some issues may be overlooked or underreported in the early design stage, such as patient discomfort, complexity of use, or increased weight and thickness of the insole. In addition, the real-life experiences of patients may not be fully captured, which limits the generalizability of the results. To reduce these risks, it is important to consider critical expert opinions and to discuss realistic usage scenarios during the interviews.

7. Timeline

Table 1. Research Stages Timeline in Spring 2026

Task / Month(2026)	January	February	March	April	May	June
Obtaining Research Permit						
Sending interview invitations and scheduling sessions						
Conducting interviews						
Categorizing the data						
Conclusion and Summarization						

Appendix 2 - Privacy Notice

Title of the Study:
Design and Implementation of a Smart Insole for the Prevention of Diabetic Foot Ulcers.

Researcher.....,
Master's student of Information Technology in the field of Medical Technologies,Metropolia University of Applied Sciences, Finland.

Supervisor:.....

Purpose of the Study
The purpose of this study is to explore the concept of a wearable medical device designed to measure plantar pressure and help prevent diabetic foot ulcers using a smart insole with offloading capability.

Data Controller
The research is conducted as part of a Master's thesis atMetropolia University of Applied Sciences.
The researcher is responsible for collecting and processing the data under the supervision of the thesis supervisor.

Legal Basis for Processing Personal Data
The legal basis for processing personal data in this research is voluntary consent from the participants.

The following data may be collected:
Professional background related to the research topic
Opinions and experiences shared during the interview
Voice recordings of the interview (if the participant agrees)

No sensitive personal data such as identification numbers, medical records, or financial information will be collected.

Data Storage and Protection

All collected data will be handled confidentially.

Participants will be identified using participant codes (e.g., P1, P2).

Personal identifiers will not be included in research results or publications.

Data will be stored securely on password-protected storage accessible only to the researcher and the supervisor.

Data Retention

Voice recordings will be deleted after transcription and verification.

Research data will be retained only as long as necessary for the completion of the thesis and will then be securely deleted.

Sharing of Data

Research results may be published in the thesis or academic publications.

However, all results will be presented in anonymized form, and individual participants cannot be identified.

Participants' Rights

Participants have the right to:

receive information about the research

withdraw from the study at any time without consequences

withdraw their consent for data processing

request deletion of their personal data where applicable

Contact Information

If you have questions about the study or the processing of your personal data, please contact:

Researcher:.....

Email: .....

Phone: ……

Appendix 3 - Consent to Participate in Research

Title of the Study:
Design and Implementation of a Smart Insole for the Prevention of Diabetic Foot Ulcers.

Researcher.....,
Master's student of Information Technology in the field of Medical Technologies,Metropolia University of Applied Sciences, Finland.
Phone: .....
Email: .....

Supervisor:
.....

I have been invited to take part in the above-mentioned research study. The purpose of this study is to explore the idea of a wearable medical device. The goal is to find a way to measure plantar (foot sole) pressure and help prevent diabetic foot ulcers (DFU) using a smart insole with offloading capability.

The researcher contacted me because my educational background and professional experience are relevant to this topic and I have been informed that only the information I voluntarily share during the interview or conversation will be collected.

I understand that no sensitive personal data (such as identification numbers, medical records, or financial information) will be collected. All collected information will be stored anonymously using participant codes (P1, P2, etc.).

I understand that I can withdraw from the study or withdraw my consent at any time, without giving any reason and without any negative consequences. If I

withdraw, the data collected up to that point may still be used in the study. If I withdraw my consent, any personal data collected about me will no longer be used and will be securely deleted, as long as deletion is still possible.

I have received enough information about the study and how my data will be handled. I have had the opportunity to ask questions, and I have had enough time to decide whether I want to participate.
By signing below, I voluntarily agree to participate in this research study.

I have been informed that the data will be collected only through direct interviews with participants. No external sources will be used, and my information will not be shared with any individual, company, or organization outside the research team without my written consent.

Any benefits that I may receive from participating in this interview have not influenced my decision to participate in this research, and I confirm that my participation is entirely voluntary.

Consent Options

I agree that my voice may be recorded during an in-person interview, or that the conversation may be recorded if the interview is conducted online. I have been informed that all interview data will be securely deleted three months after the completion of the thesis and research.

☐ Yes

☐ No

I agree that anonymized research data may be stored and made available for use by other researchers.

☐ Yes

☐ No

If a prototype of the device is available, I voluntarily agree to test its functionality at my own responsibility. I understand that the device is a research prototype, has not been clinically approved, and is not intended for medical diagnosis or treatment. I acknowledge that my participation in testing the prototype is entirely optional.

☐ Yes

☐ No

If consent is the legal basis for processing personal data in this study, I also confirm that I agree to the processing of my personal data. I understand that I have the right to withdraw my consent as described in the privacy notice.

Date:.....……

Participant's Name:

.....

Participant's Signature:

…….....……

Date:….....……

Researcher's Name:

.....

Researcher's Signature:

…….....……

Note: This consent form (two pages) is prepared in two copies. One copy will be kept by the researcher for record and archival purposes, and the other copy will be provided to the participant.

Appendix 4 - Information Sheet for Participants

Title of the study
Design and Implementation of a Smart Insole for the Prevention of Diabetic Foot Ulcers.

Invitation to participate
You are invited to participate in a study aimed at exploring the idea of a wearable medical device. The goal of the study is to find a way to measure plantar (foot sole) pressure and help prevent diabetic foot ulcers (DFU) using a smart insole with offloading capability.

We have identified you as a suitable participant because you have experience or expertise in orthotics, smart insoles, or caring for patients with diabetic foot ulcers, whether as a doctor, lecturer, staff member, student, or other relevant professional.

Please read this information sheet carefully. If you have any questions about the study, you can contact the responsible researcher (contact details are at the end of this document). If you decide to participate, you will be asked to give a separate consent.

Voluntary participation
Participation in this study is voluntary. You may refuse to participate, withdraw consent at any time, or cancel your consent without giving a reason. Your decision will not affect your studies, work.

You have the right to withdraw your consent for the processing of your personal data at any time. The withdrawal will take effect on the day it is made. As a rule, this does not affect the use of data collected before the withdrawal.

Purpose of the study



The purpose of this study is to develop a wearable medical device to help prevent diabetic foot ulcers. The device is a smart insole that fits inside the shoe and uses pressure sensors to measure the pressure on different areas of the foot, what makes this idea unique and different from existing products is its offloading feature, which addresses a gap in current research. We would like to consult with you to get your opinions on the concept, including its potential challenges, advantages, and drawbacks.

Researchers and organizers

This study is conducted by Arash Ahmadinia, Master's student in Information Technology, under the supervision of Sakari Lukkarinen, Senior Lecturer atMetropolia University of Applied Sciences.

Methods and procedures

The study will be conducted through a semi-structured interview, which can be held either in person or online, depending on your preference for time and location. The interview will last approximately 1 hours to 1.5 hours.

During the interview, the idea will first be presented via a video demonstration, showing how the device works. The discussion will then focus on the concept and its effectiveness in preventing diabetic foot ulcers. Your feedback on the design, usability, practicality, advantages, disadvantages, and other aspects of the prototype will be extremely valuable.

Possible benefits and risks

The results of this study are expected to have benefits for the prevention of diabetic foot ulcers, and we hope that the outcomes of this thesis will contribute to advancing knowledge in wearable medical devices. By drawing on your

experience and expertise, we aim to bring this idea closer to practical implementation.

At the same time, careful attention must be paid to the design, material, and specifications of the insole, as insoles can affect walking patterns and movement over the long term. Potential risks for patients and users must also be carefully considered.

The interview has no risk for the interviewee.

Costs and compensation

Participation is free of charge. No separate compensation will be paid for participation.

Data storage and reporting of results

The research will start on 01.02.2026 and continue until 31.08.2026. All collected data will be stored securely for three months after the end of the project for final analysis and verification, and will then be safely deleted and after this retention period, all digital files will be permanently erased, and any printed materials will be destroyed.

Data analysis will only be carried out by the researcher and the academic supervisor. No other individuals or organizations outside the research team will have access to the data.

The data will be collected only through direct interviews with participants, no external sources will be used and your information will not be shared with any person, company, or organization outside the research team without your written consent.

No sensitive personal data (such as identification numbers, medical records, or financial details) will be collected. All collected information will be stored anonymously using participant codes (P1, P2, …) and the results will be

presented only in summarized form, without mentioning any names or personal details and they may appear in a university report, thesis, and, if possible, in scientific publications.

End of the study
31.08.2026

If the conditions for data collection for the thesis change, the researcher may need to modify or terminate the study accordingly.

Further information

If you have any questions about the study or wish to withdraw your participation, please contact the researcher:
……..

Contact details of researchers

Thesis Author
Name…….
Phone number: …….
Email: …….
Organization : ……

Thesis supervisor
Name: ……..
Phone number: ……..
Email: ……..
Organization : ……

Appendix 5 - Data Management Plan created using DMPTuuli

Title: MSOLE-Design and Implementation of a Smart Insole for the Prevention of Diabetic Foot Ulcers

Creator: Arash Ahmadinia

Principal Investigator: Arash Ahmadinia

Contributor: Sakari Lukkarinen

Affiliation:Metropolia University of Applied Sciences

Template: DMP for Thesis

Project abstract:
In this thesis, I focus on exploring a new way to design and improve smart insoles that can help prevent or treat diabetic foot ulcers. The approach considers both patient needs and what has already been studied or developed in the market. The work starts with a look at the background and previous efforts, then moves on to defining a clear research question and suggesting a possible solution. Finally, it presents the results, discusses the benefits and practical value of the proposed design, and considers the challenges that come with it.

Design and Implementation of a Smart Insole for the Prevention of Diabetic Foot Ulcers

1. General description of the data

1.1. Briefly describe the data you collect and/or produce or that already exists, as well as their properties (type, file format, size, access rights, collection methods). Create a table or list of the data.

Data type	Description	File format	Collection method	Access rights
Audio interview	Face-to-face interview recordings	WAV	Self-collected	Researcher only
Video interview	Online interview recordings	MP4	Self-collected	Researcher only
Interview framework	Interview questions and structure	PDF	Self-produced	Researcher only

1.2. How do you ensure the consistency and accuracy of the data?

To ensure that the data is not accidentally changed and that the original content is preserved, all original files will be stored securely on the researcher's personal laptop, with an additional backup copy saved on an external hard drive. Before any analysis, editing, transcription, or file conversion, a copy of the original data will be created and kept unchanged. Version control will be used by systematically naming files (e.g. V1, V2, V3) to track different stages of processing. Clear file naming conventions and a structured folder system will be applied throughout the thesis process. Data quality will be maintained by carefully managing audio and video recordings, checking files after transfers or conversions, and reviewing transcriptions to avoid errors. These practices help ensure that the data remains accurate and consistent throughout its lifecycle.

2. Personal data, ethical principles and legislation

2.1. Does the data contain any personal information? If yes, enter in the supplementary information field: the personal data to be collected, whether there are special categories of personal data.
No

2.2. Who has the main responsibility for the processing of personal data, i.e. controllership? If you do not collect personal data, you can skip this question.
No

2.3. What measures are required to ensure data protection in my thesis? If you do not collect personal data, you can skip this question.

This research does not collect, store, or process any personal or identifiable data. Participants are informed in advance that no personal data will be used and that the interviews are conducted anonymously. Participation is voluntary, and consent to participate in the research is obtained separately.

Privacy policy/statement: Not applicable to my research
Consent for processing personal data: Does not apply to my research
Research bulletin: Not applicable to my research

2.4. Do you need an ethical review? To the additional information field: justification and possible implementation

I have reviewed the ethical review guidelines and criteria for ethical review. The research design does not involve any deviation from the principle of informed consent, does not interfere with the physical integrity of the participants, does not involve minors, and does not expose participants to strong stimuli or situations that could cause psychological harm beyond normal everyday life. The research does not pose any security risks to the participants or the researcher.
Potential risks are minimal and limited to ordinary interview-related discomfort. Any possible risks will be clearly explained to the participants in the consent form, and participation is entirely voluntary.

2.5. Are there other research ethical questions related to the data?

No.

2.6. How will you manage the rights to the data you use, produce and share?

The student retains full ownership of the research data. The data is used only for thesis purposes and is not transferred or shared with third parties. The supervisor only oversees the academic quality of the work.

3. Data description and documentation

3.1. How do you describe and document the data in an understandable way?

Microsoft Excel 365 will be used to store, organize, and analyze the collected data. It enables basic descriptive statistical analysis to summarize the data and identify general patterns and Power Query, integrated within Excel 365, will be used for data cleaning and transformation. It ensures a consistent, accurate, and repeatable data preparation process with automatic updates.

4. Storage and data security during the thesis process

4.1. Describe here where the data will be stored and how it will be backed up during the thesis process.

Since the research does not involve sensitive or personal data, all data will be stored on the researcher’s personal laptop. An additional backup copy will be kept on an external hard drive. The storage capacity of both devices is sufficient for the data volume. All files will be systematically named and organized in a structured folder system to ensure easy access, clarity, and proper documentation.

4.2. Who has access to your data, what can those people do with the data, and how do you ensure the safe transfer of the data to your potential collaborators?

The research data will be processed solely by the researcher. No one else will have access to the raw data during the thesis process. The supervisor’s role is

limited to overseeing the results and outputs of the data analysis, without direct access to the original data. Other individuals will not have access to the data until the thesis is completed. All data files will be securely deleted within three months after the thesis is published.

5. Data after the thesis is completed: preserving, destruction or possible further use and opening

5.1. Is the data or part of it preserved? If yes: describe which data or part of it.
No
5.2. Will the data be destroyed or part of it? If yes, describe which data or part of the data, and how and when the destruction will occur.
Yes
I am aware of the requirements for data destruction. All interview data collected for this thesis will be securely deleted three months after the thesis is published. The destruction will include all digital files, which will be permanently removed from both the researcher's personal laptop and backup external hard drive.

5.3. Are you planning to further use or open your data or part of it? If yes, describe which data or part of it. Also describe whether the data will be opened for public use with a license, or whether it will be handed over to a specific party for further use, as well as the possible measures that opening the data or preparing for further use requires (for example, anonymisation, permissions from research subjects, agreements with collaborators).

Yes, the research data will not be shared publicly in its raw form. However, the thesis will be published in the Theseus. All references to individual participants will be anonymised and coded using identifiers such as P1, P2, P3, etc. No personal information of the interviewees will be disclosed.
Participants will only be categorised broadly based on their role or profession (for example, researcher, teacher, or staff member) to provide contextual information without revealing identities. This ensures that the data cannot be traced back to individual participants.

6. Responsibilities and resources

6.1. Who is responsible for data management and what kind of resources does data management require?

The researcher is responsible for all aspects of the research process, including data collection, ensuring data quality, data storage and backup, data security, and data protection. The researcher also manages any steps related to preparing the data for the thesis, including anonymization and coding.